

Automated Generation and Operation of Digital Twins for Manufacturing Systems



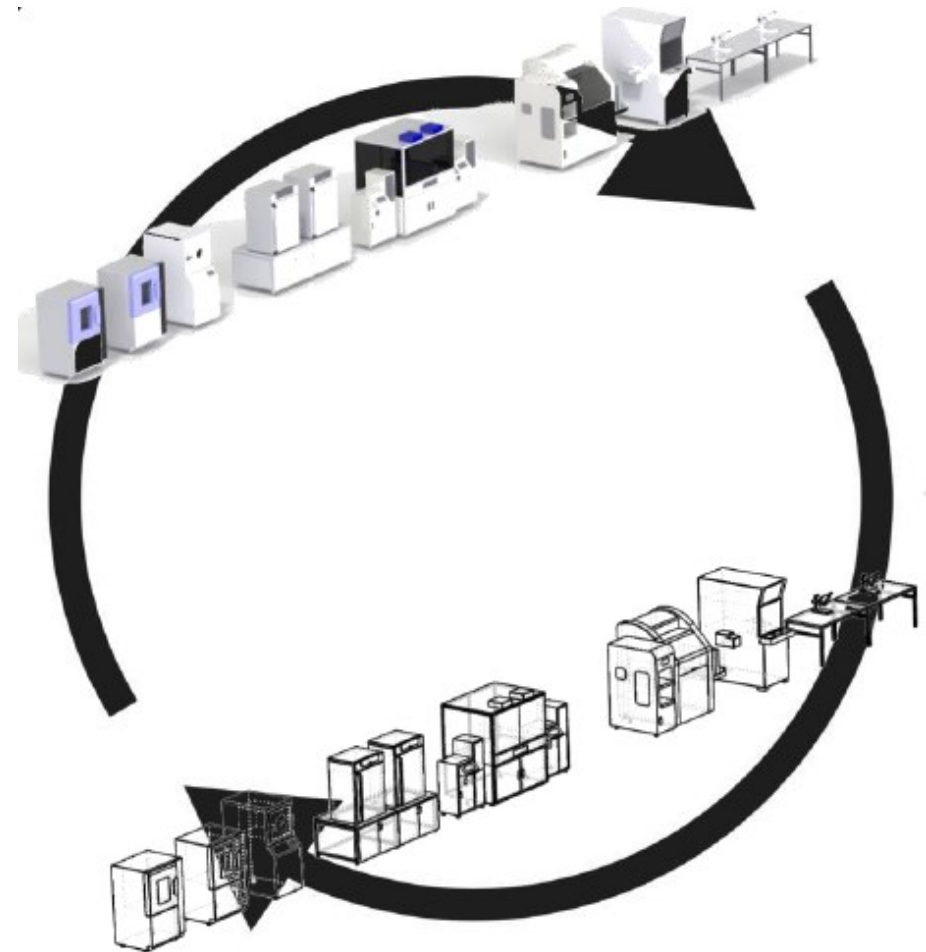
Giovanni Lugaresi

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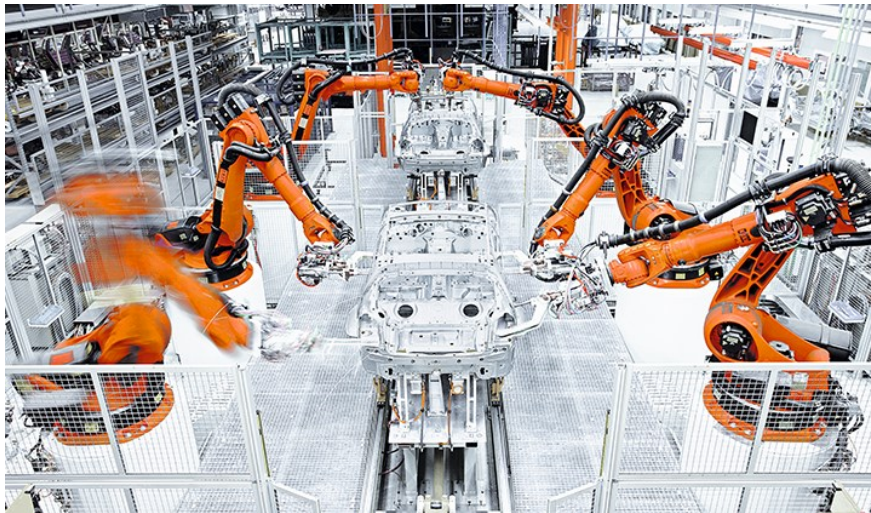
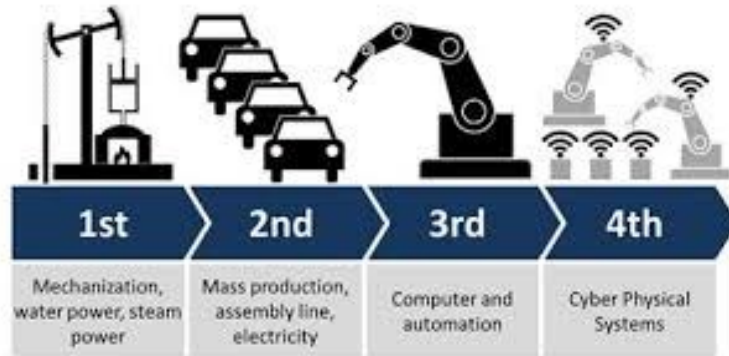
November 17th, 2023

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Background



CHALLENGES

- Unpredictable market demand
- Complexity
- Pressures on cost reduction
- High-level of automation
- Failures impact

OPPORTUNITIES

- ✓ High frequency data collection
- ✓ Understanding emerging behaviors
- ✓ Evaluating alternative scenarios
- ✓ Affordable data analytics

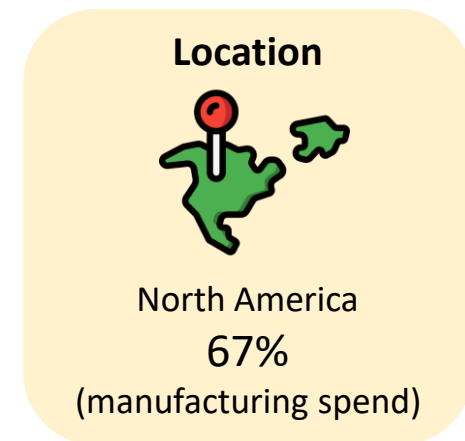
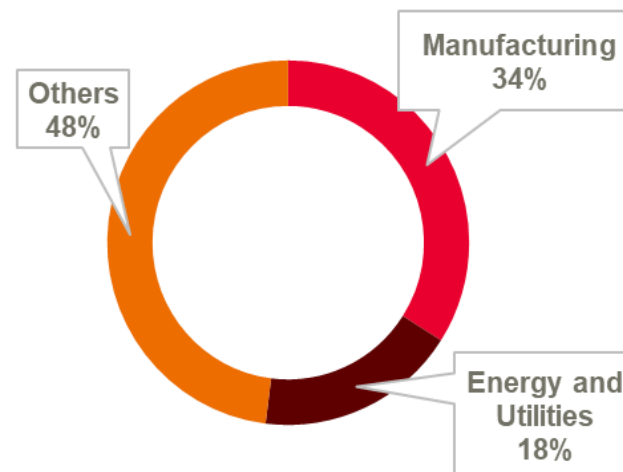
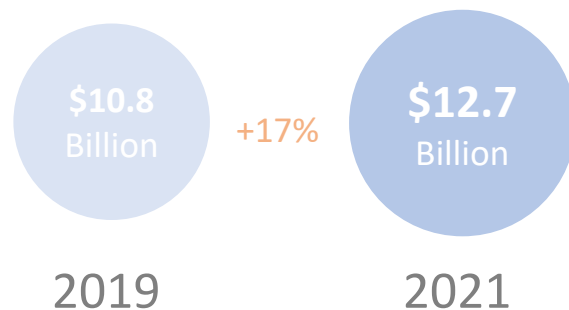


DIGITAL SUPPORT
TOOLS FOR
PRODUCTION
PLANNING AND
CONTROL

Background

- **Manufacturing** is the biggest sector for Digital Twins deployment, accounting for 34% of total spend in 2021¹.
- The global **Digital Twin market size** was valued 8.6 billion USD in 2022 and it is expected to reach 137.7 billion USD by 2030 with a CAGR of 42.6%².
- Only around 5% of companies affirm DTs are not part of their **digital transformation strategy** whereas another 86% consider DTs a crucial solution in their strategy³.

Global Expenditure on Digital Twin



Sources: (1) Juniper. (2) Fortune Business Insights. (3) Dertien and Macmahon 2022.

From Simulation into Digital Twins

Time	1960 – 1979 (Tocher, 1967)
Objects	Digital Model (Simulation)
Interaction with Physical system	▪ Offline ▪ Physical to digital
Main Services	▪ Design ▪ Off-line Analysis
Examples of decisions	▪ Assembly Line Balancing ▪ Layout Optimization ▪ Resource dimensioning ▪ etc

Kritzinger, W., Karner, M., Traar, G., Henjes, J., & Sihn, W. (2018). Digital Twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11), 1016-1022.

From Simulation into Digital Twins

Time	1960 – 1979 (Tocher, 1967)	1980 – 2010 (Williams, 1989)
Objects	Digital Model (Simulation)	Digital Shadow (Mirror)
Interaction with Physical system	▪ Offline ▪ Physical to digital	▪ On-line ▪ Physical to digital
Main Services	▪ Design ▪ Off-line Analysis	▪ Monitoring ▪ Scenario Evaluation ▪ Prediction
Examples of decisions	▪ Assembly Line Balancing ▪ Layout Optimization ▪ Resource dimensioning ▪ etc	▪ Scheduling ▪ Due-date quotation ▪ Material Resource Planning ▪ etc

Kritzinger, W., Karner, M., Traar, G., Henjes, J., & Sihn, W. (2018). Digital Twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11), 1016-1022.

From Simulation into Digital Twins

Time	1960 – 1979 (Tocher, 1967)	1980 – 2010 (Williams, 1989)	2011 – (Industry 4.0)
Objects	Digital Model (Simulation)	Digital Shadow (Mirror)	Digital Twin
Interaction with Physical system	- Offline - Physical to digital	- On-line - Physical to digital	- On-line - Bidirectional
Main Services	- Design - Off-line Analysis	- Monitoring - Scenario Evaluation - Prediction	- Optimization - Control - Coordination
Examples of decisions	- Assembly Line Balancing - Layout Optimization - Resource dimensioning - etc	- Scheduling - Due-date quotation - Material Resource Planning - etc	- Re-Scheduling - Dispatching - Maintenance planning - etc

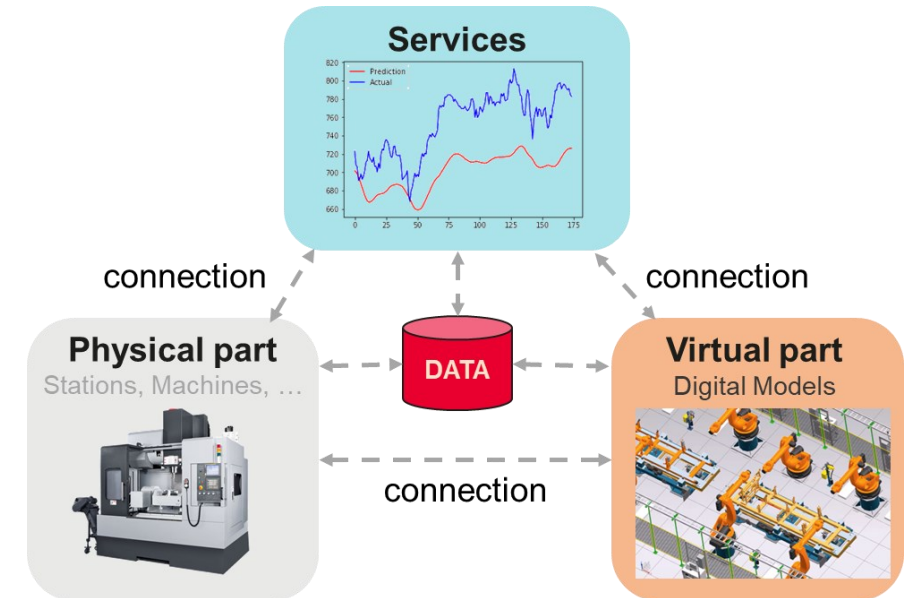
Kritzinger, W., Karner, M., Traar, G., Henjes, J., & Sihn, W. (2018). Digital Twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11), 1016-1022.

Background

Digital Twins for Manufacturing Systems

Five-dimension Model:

- *Physical part*
- *Virtual part*
- *Connection*
- *Data*
- *Services*



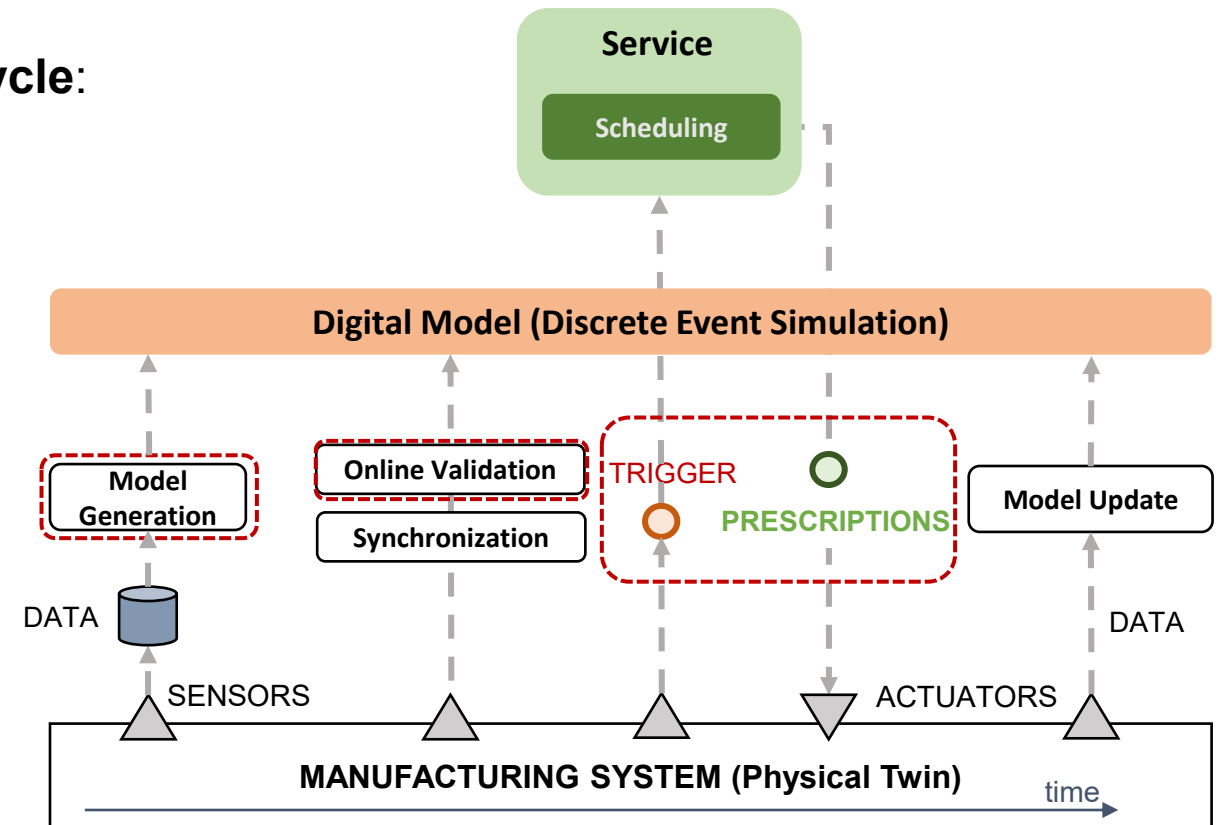
Physical Entity	Virtual Entity	Services	Data	Connection
Manufacturing Systems: • Electric Devices • Semiconductor • Automobile Components • CNC Machines • Conveyors	• Data Models • Analytical Models • Surrogate Models • Simulation Models • Physical Models	• Prediction • Scheduling • Sequencing • Dispatching • Re-scheduling • Optimization • Performance Evaluation • Anomaly Detection	• Time-series Databases • Relational Databases • Cloud Services	• Standards (OPC, Profinet, etc.) • Internet-of-Things (MQTT,...)

Tao, F., Zhang, H., Liu, A., & Nee, A. Y. (2018). Digital twin in industry: State-of-the-art. *IEEE Transactions on industrial informatics*, 15(4), 2405-2415.

Background

Digital Twins may be subject to the following **life cycle**:

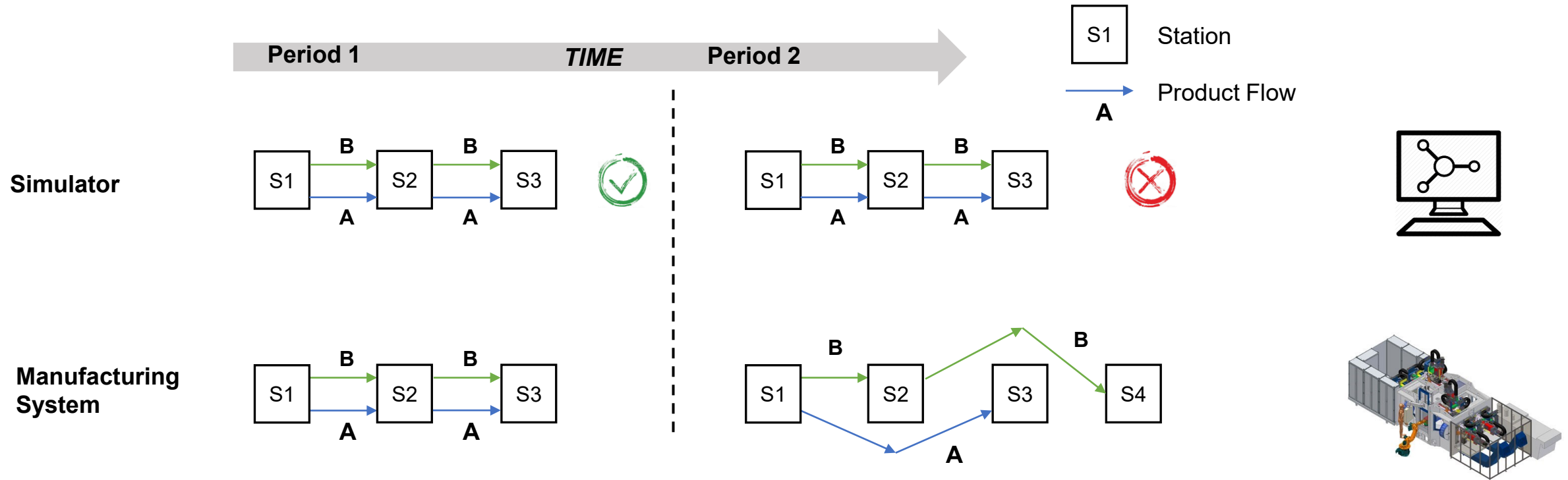
- Being **generated** from available data.
- **Being used** to provide a particular *service*.
 - The model must be *synchronized* to the real system state.
 - The model must be *valid*.
- Being **updated** to adapt to the changes of the physical system.



The results of the analysis must be used to determine **actionable commands** on the real system.

Problem Introduction

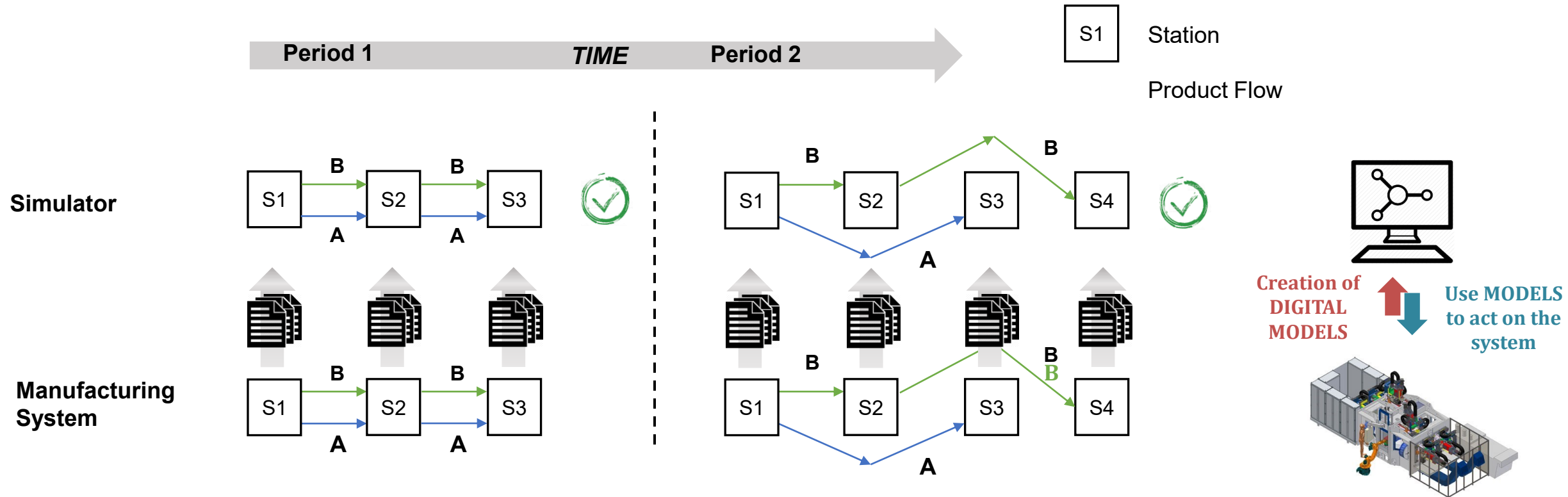
At the time the physical manufacturing system changes, the simulator might be out of date.



Manufacturing systems change frequently due to external drivers (e.g. demand, price uncertainty). Hence, **current simulation techniques are poor** as tools for **short-term decision making**.

Problem Introduction

At the time the physical manufacturing system changes, the simulator might be out of date.

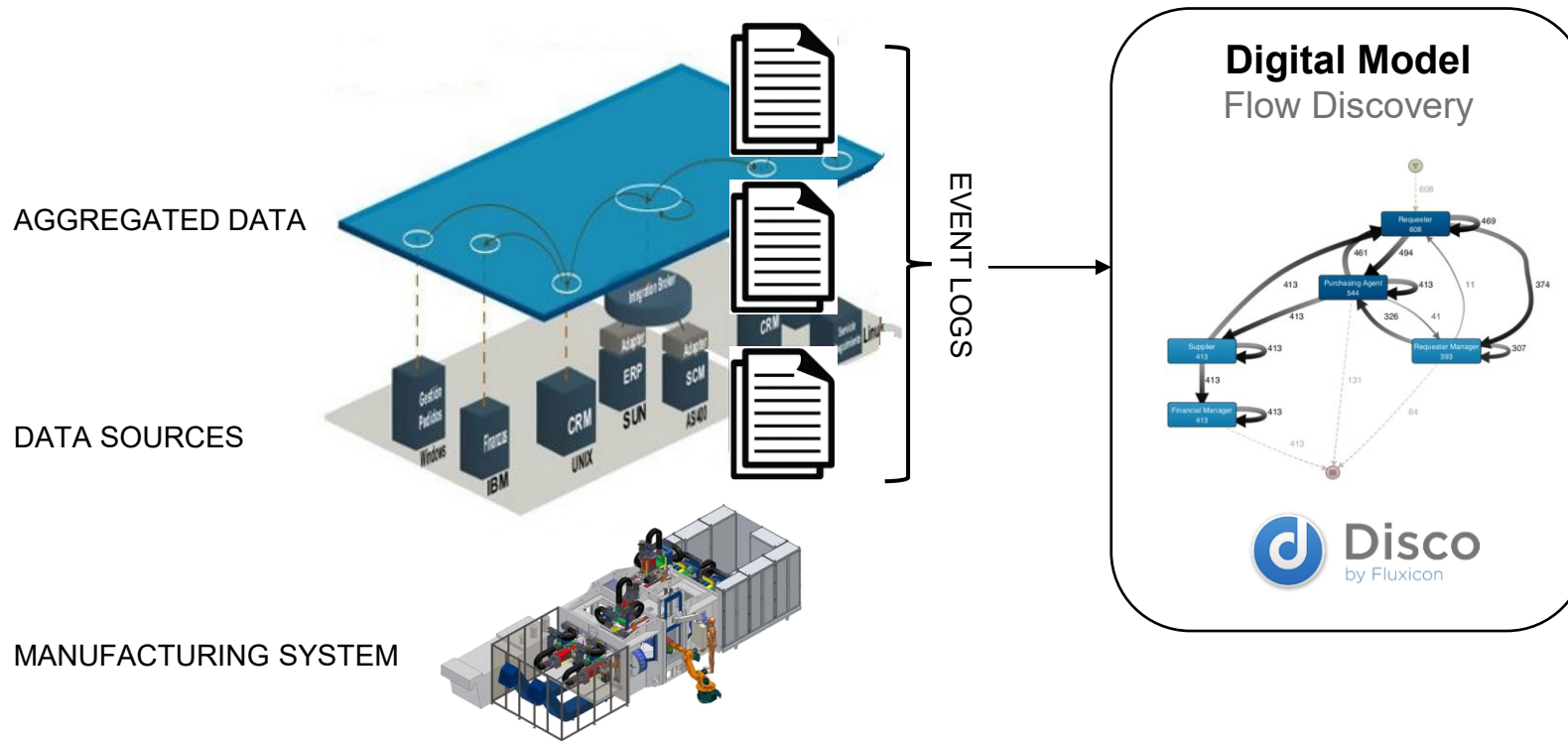


By exploiting the data produced by the parts and resources, it is possible to achieve higher reactivity in the simulation model building and update phase.

Problem Introduction

Lugaresi, G., & Matta, A. (2020). Generation and tuning of discrete event simulation models for manufacturing applications. 2020 Winter Simulation Conference (pp. 2707-2718). IEEE.

- **Process mining** is a novel discipline that is concerned with discovering and deriving useful insights from operational process execution logs
- Applications: health care, ICT, service processes, business processes, transportation.



Limitations for manufacturing systems

- Finite capacity resources cannot be recognized automatically.
- User is not free in the choice of **aggregation level**.
- Highly sensitive to **rare or wrong sequences** of events.
- No **relationship with performance** estimation from the obtained model.
- Absence of model generation framework.
- Absence of data logging guidelines.

Van Der Aalst, Wil. "Data science in action." *Process mining*. Springer, Berlin, Heidelberg, 2016. 3-23.

Model Generation

Lugaresi, Giovanni, and Andrea Matta. "Automated manufacturing system discovery and digital twin generation." *Journal of Manufacturing Systems* 59 (2021): 51-66.

Model Generation Problem

Use material flow tracking data to generate a graph model representing activity relationships in the real manufacturing system.

Manufacturing System:



Event Log:

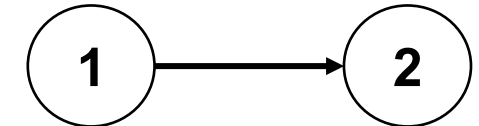
Time-stamp	Part ID	Activity
2020-11-23 16:37:40	1	Station 1
2020-11-23 16:37:44	1	Station 2
2020-11-23 16:37:47	2	Station 1
2020-11-23 16:37:51	3	Station 1
2020-11-23 16:37:52	2	Station 2
2020-11-23 16:41:02	3	Station 2

Trace of Part ID 1:
(Station 1, Station 2)

Trace of Part ID 3:
(Station 1, Station 2)

Graph Model:

"In Trace 1, Station 2 follows Station 1", ...



A graph is the mathematical representation of any simulation model with any complexity [1,2], and can be translated into simulation code [3].

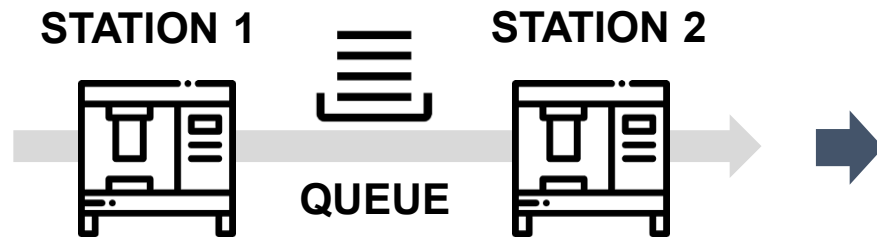
Model Generation

Lugaresi, Giovanni, and Andrea Matta. "Automated manufacturing system discovery and digital twin generation." *Journal of Manufacturing Systems* 59 (2021): 51-66.

Model Generation Problem

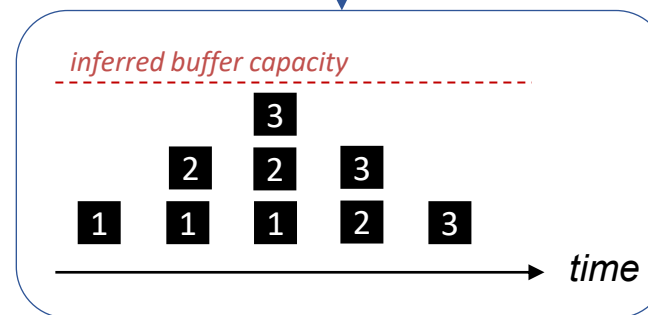
Use material flow tracking data to generate a graph model representing activity relationships in the real manufacturing system.

Manufacturing System:

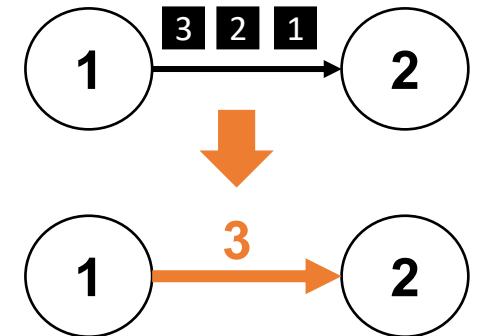


Event Log:

Time-stamp	Part ID	Activity
2020-11-23 16:37:40	1	Station 1
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2020-11-23 16:41:02	3	Station 2



Graph Model:



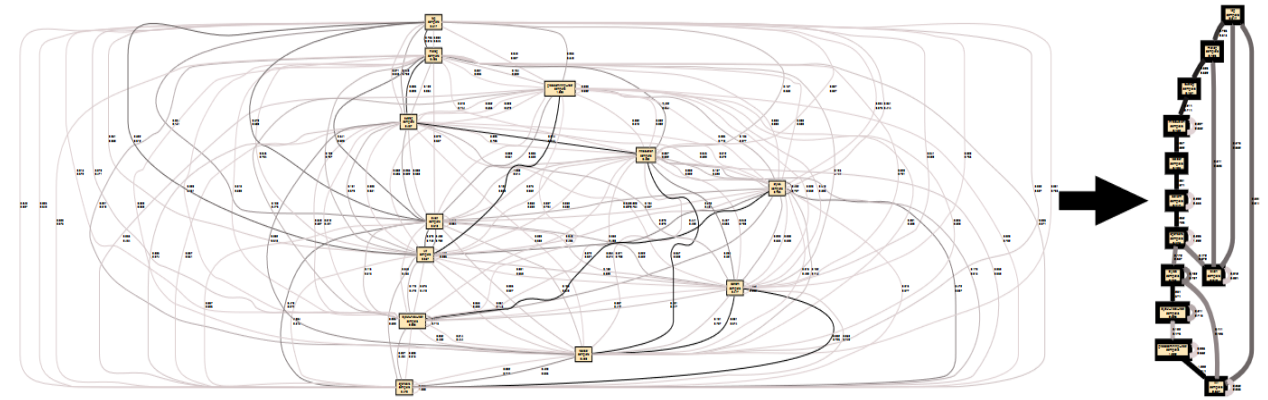
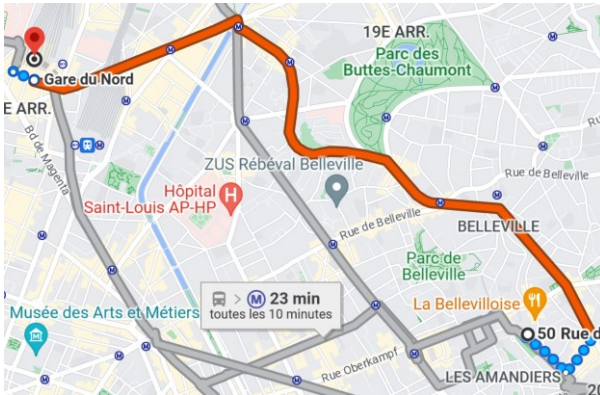
The graph is enriched with the properties of the manufacturing system: the **capacity of the arc** represents the **storage capability** between the two stations.

Model Tuning

Lugaresi, Giovanni, and Andrea Matta. "Automated manufacturing system discovery and digital twin generation." *Journal of Manufacturing Systems* 59 (2021): 51-66.

Model Tuning Problem

Process Mining may produce excessively detailed models. They work, but they are hardly useful nor human understandable. The risk is to lose sight on the modelling goal.



General Idea

Find the model with the minimum size that maximizes a score (*How well does it represent systems features?*). This corresponds to solving an optimization problem that defines nodes and edges of a graph to match an observed behavior, while guaranteeing a maximum size of the graph.

Lugaresi, G., and Matta, A.. "Automated manufacturing system discovery and digital twin generation." *Journal of Manufacturing Systems* 59 (2021): 51-66.

Günther, Christian W., and Wil MP Van Der Aalst. "Fuzzy mining—adaptive process simplification based on multi-perspective metrics." *International conference on business process management*. Springer.

Model Tuning

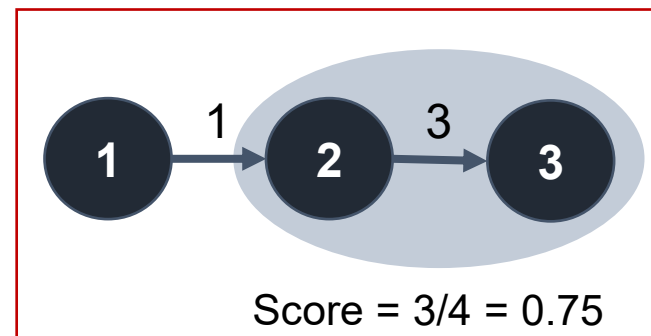
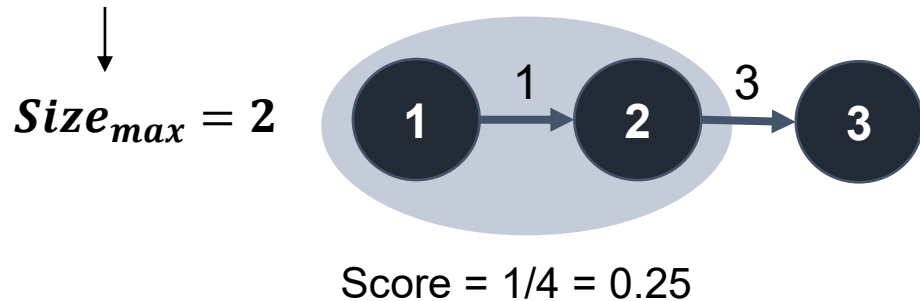
Lugaresi, Giovanni, and Andrea Matta. "Automated manufacturing system discovery and digital twin generation." Journal of Manufacturing Systems 59 (2021): 51-66.

Model Tuning Method

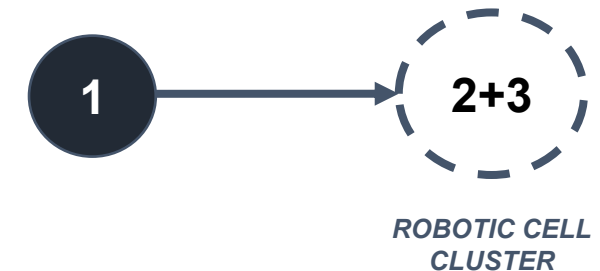
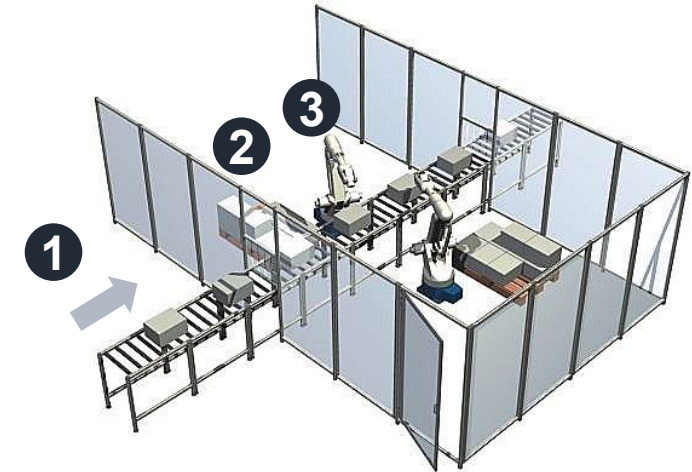
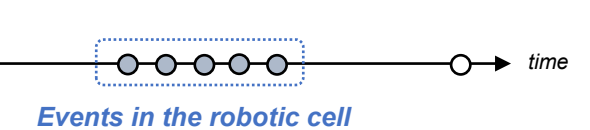
IDEA: Find the model that maximizes a score function
(How well does it represent systems features?)

$$Score_{events} = \frac{Close\ Events\ (Reduced\ Model)}{Close\ Events\ (Full\ Model)}$$

INPUT:
size of the final model

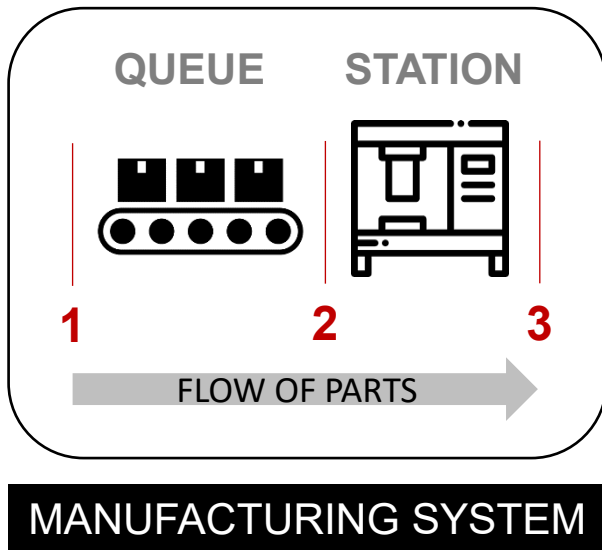


BEST ADJUSTED MODEL

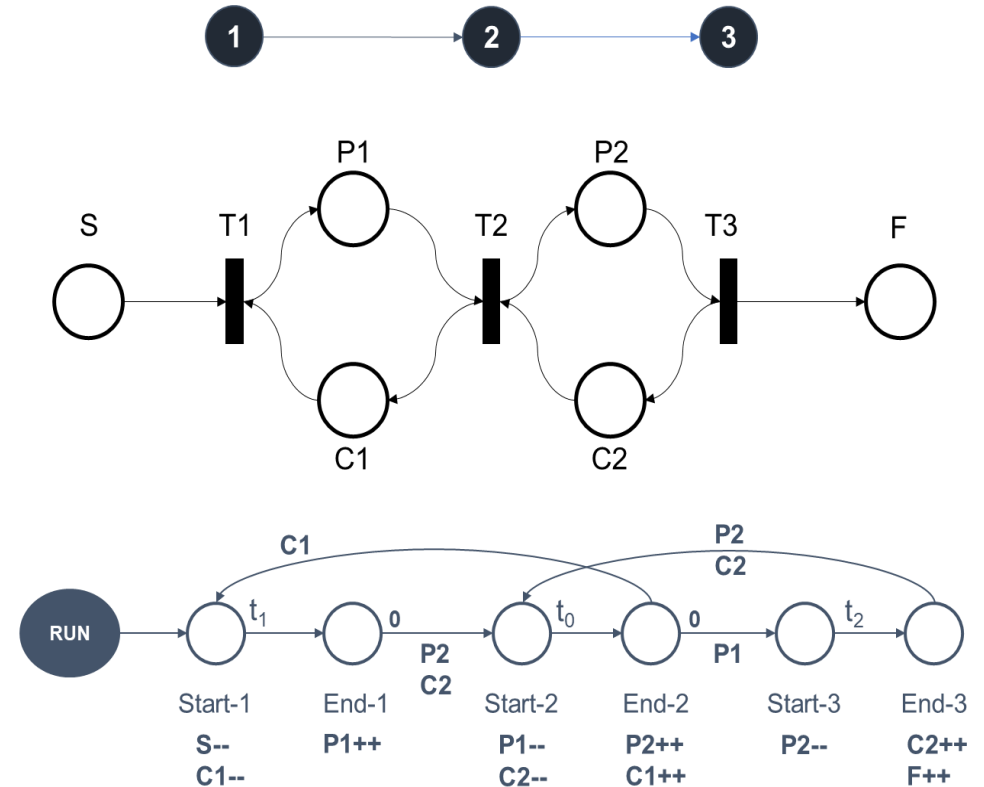


* other developed scores based on manufacturing system properties: frequency, capacity, events close in time, routing, loops.

Model Conversion



- Process model Ω_{final}
- Petri Net Model
- Event Relationship Graph*
- Commercial Simulation Software



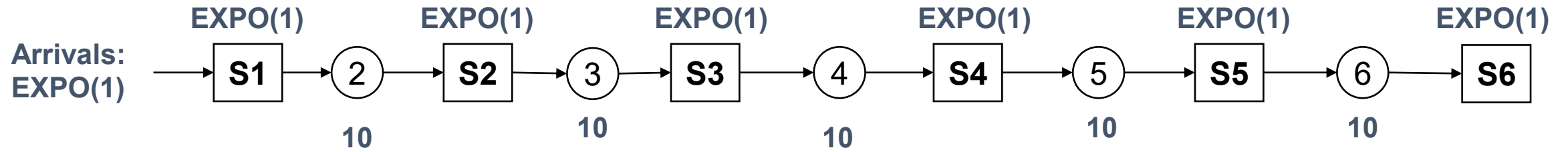
Tecnomatix Plant
Simulation



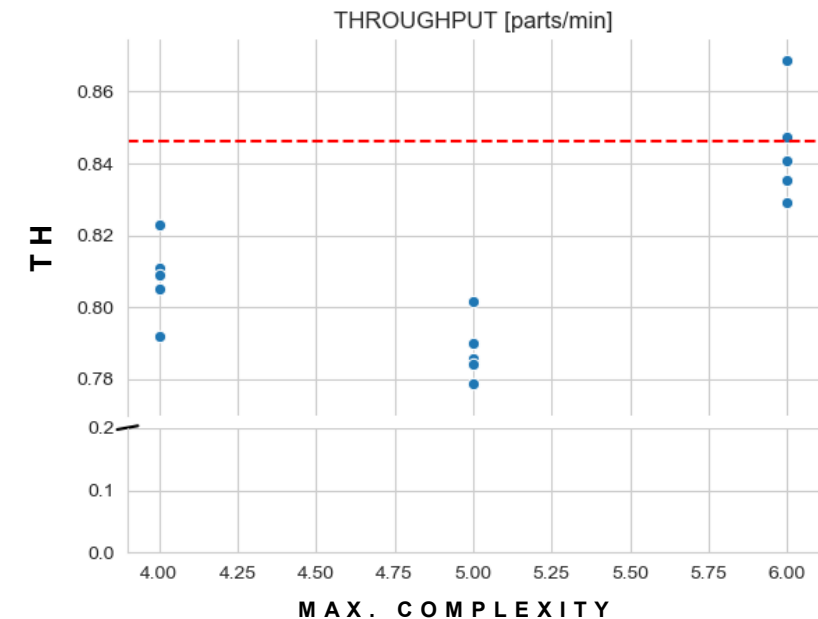
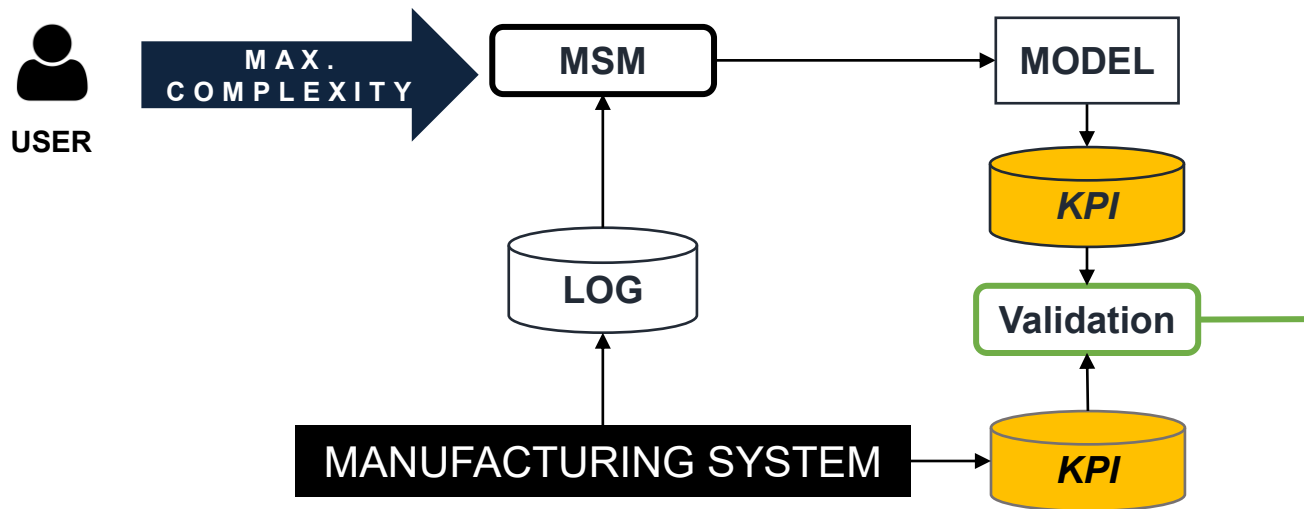
* further reduction may be applied

*Schruben, Lee, and Enver Yucesan. "Transforming Petri nets into event graph models." *Proceedings of Winter Simulation Conference*. IEEE, 1994.

Test Cases

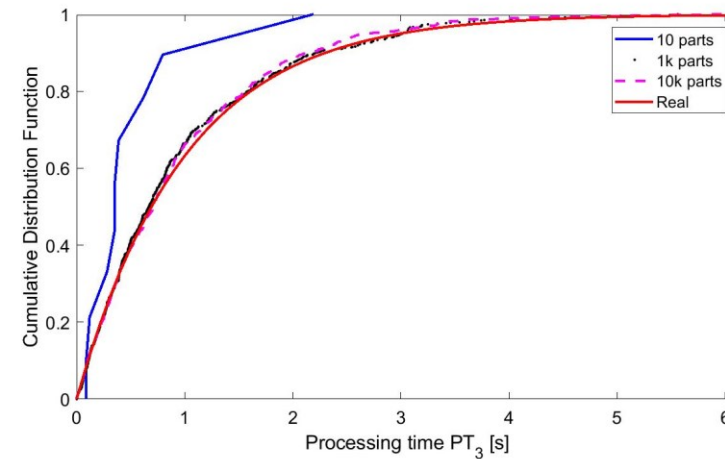
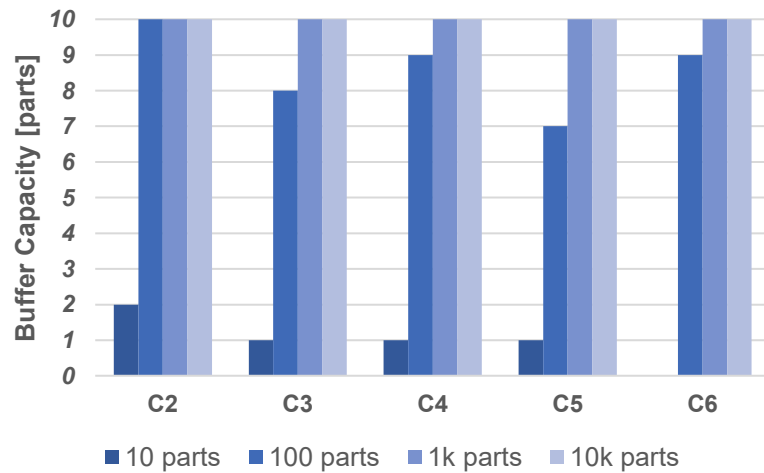
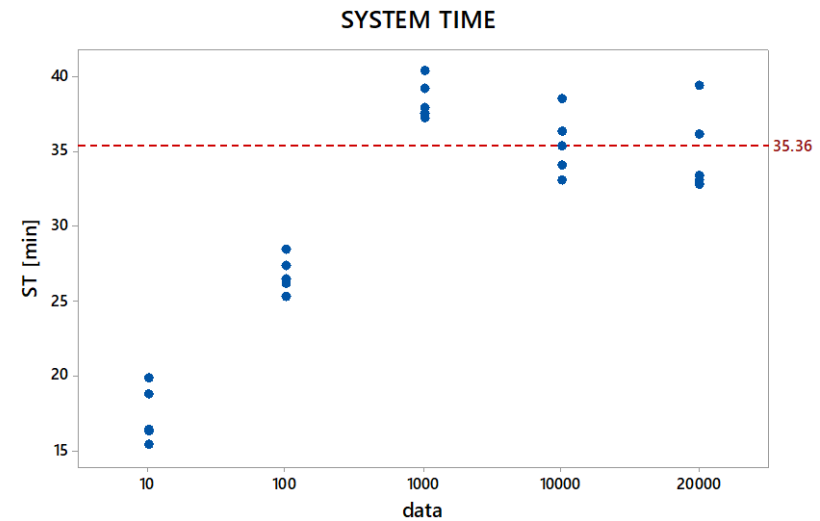
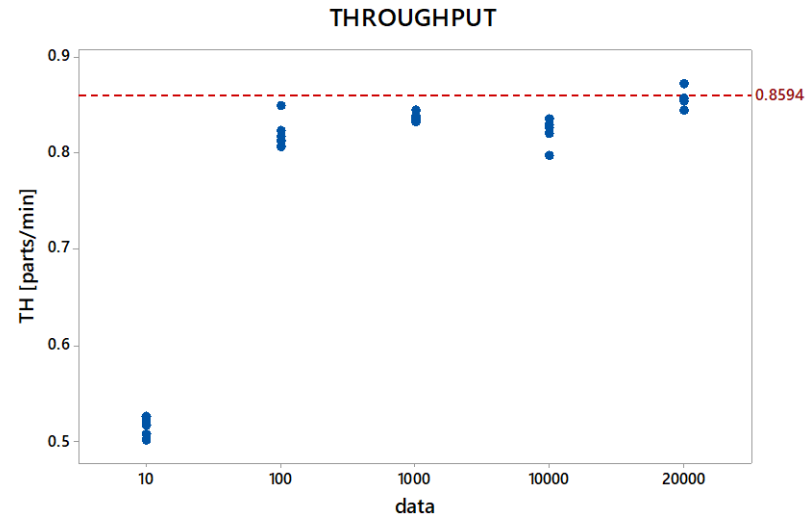


PERFORMANCE COMPARISON: AV. ERROR = 4.7 %
MAX ERROR = 11.8 %



NOTICE: Validation can also be applied online! (Online Model Adaptation)

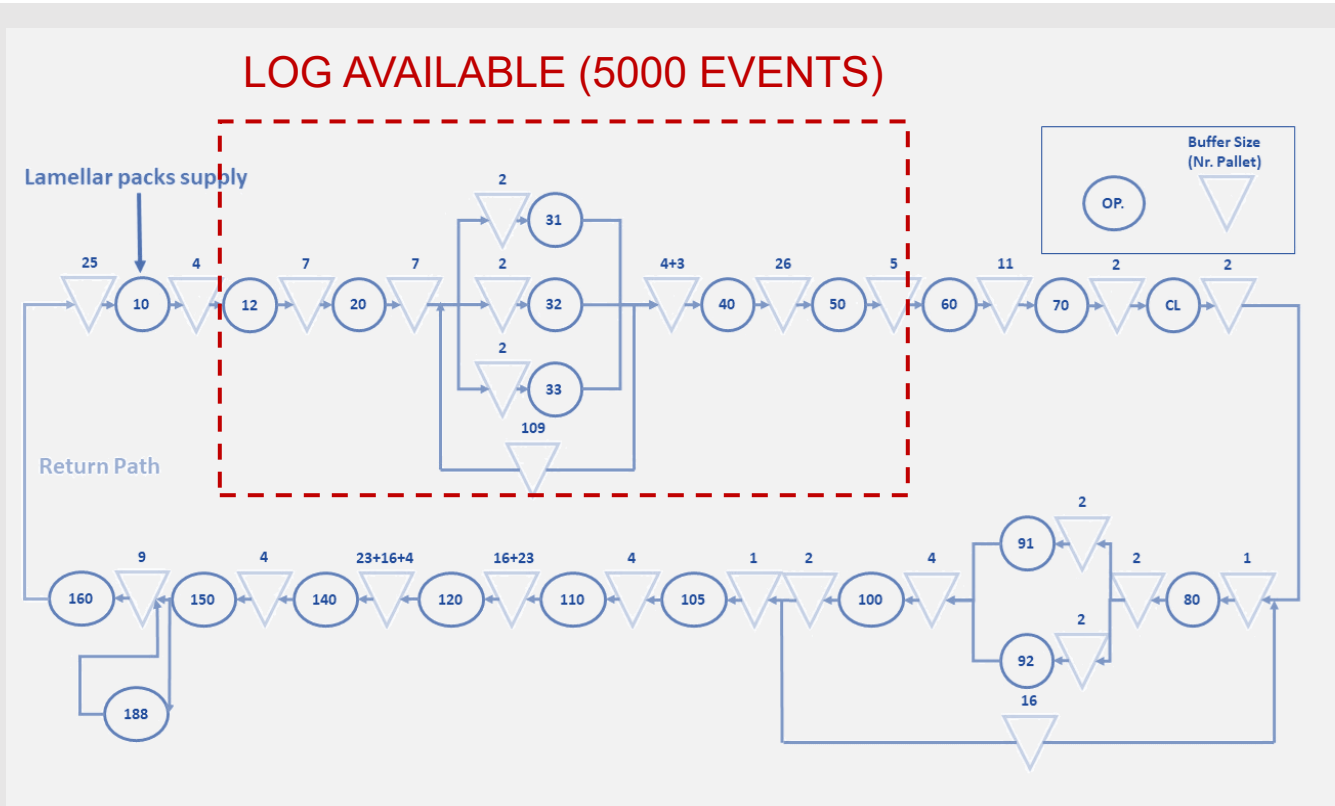
Test Cases



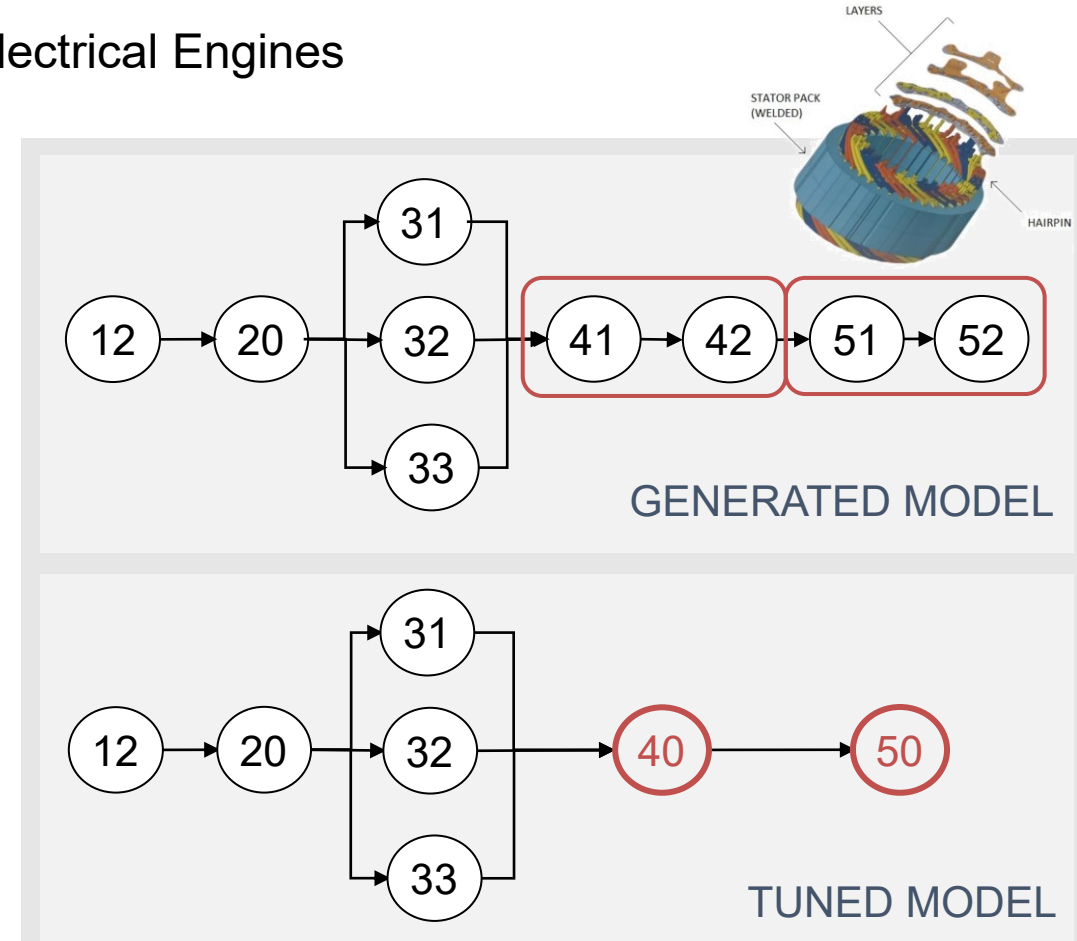
Test Cases

Lugaresi, Giovanni, and Andrea Matta. "Automated manufacturing system discovery and digital twin generation." Journal of Manufacturing Systems 59 (2021): 51-66.

Real Production Line: Closed-Loop Pallet Production Line of Electrical Engines



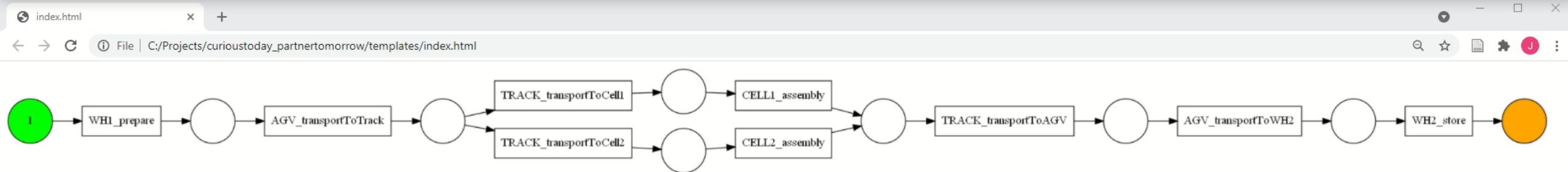
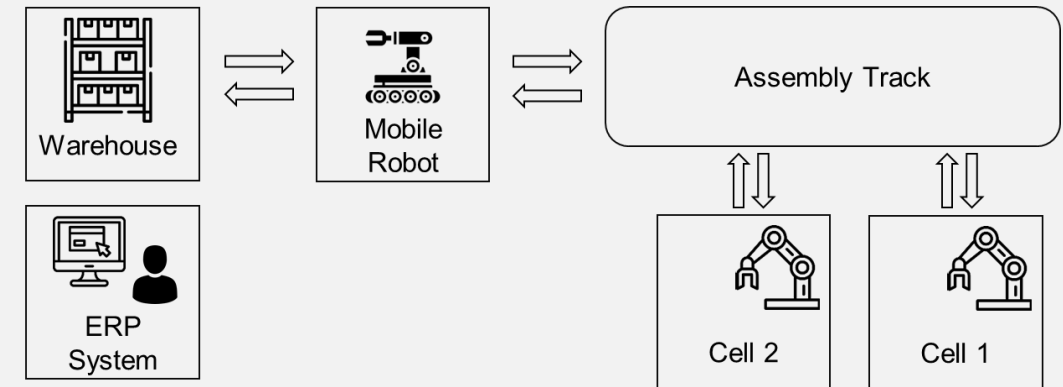
Manufacturing System (EV Stator Line)



Digital Model (in less than 30 s)

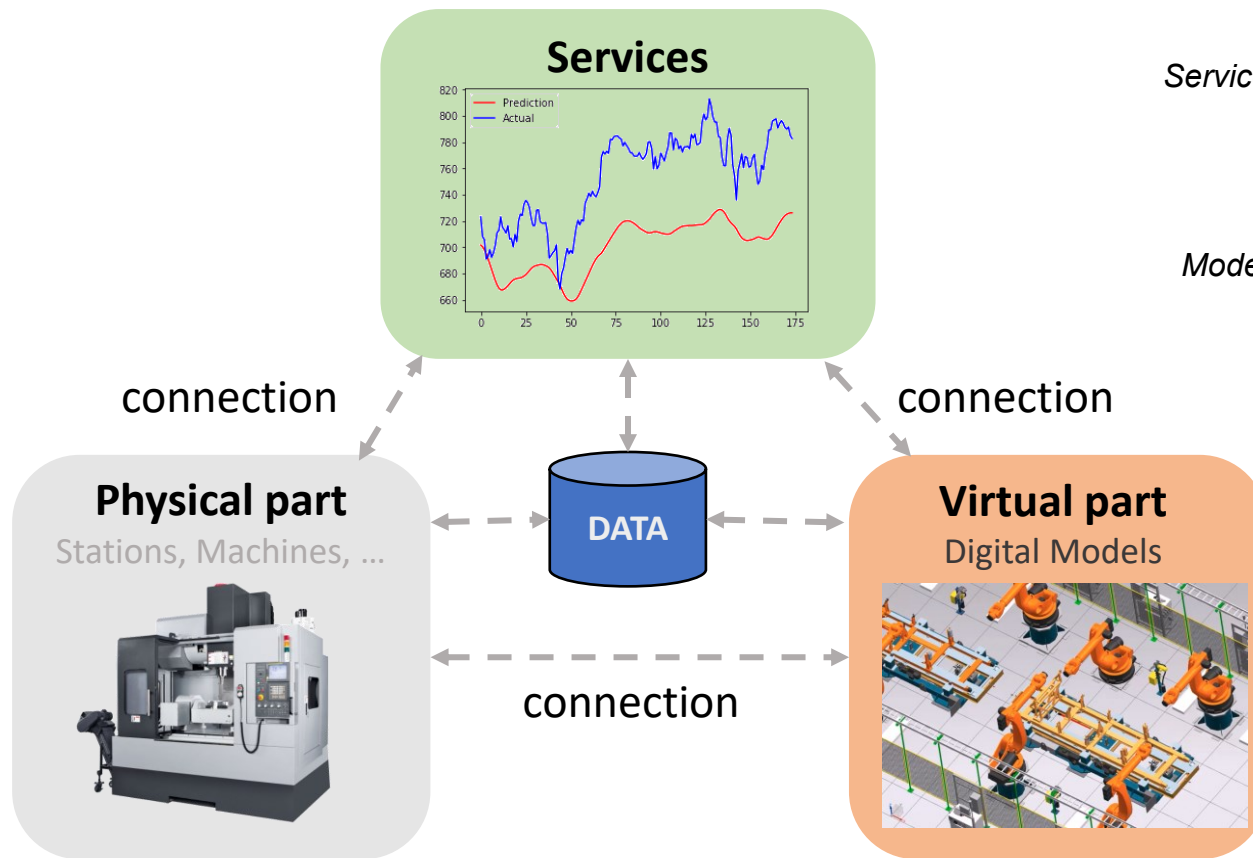
```
C:\Projects\curioustoday_partnertomorrow>python twocell.py
```

Collaboration with **SDU** (Denmark)

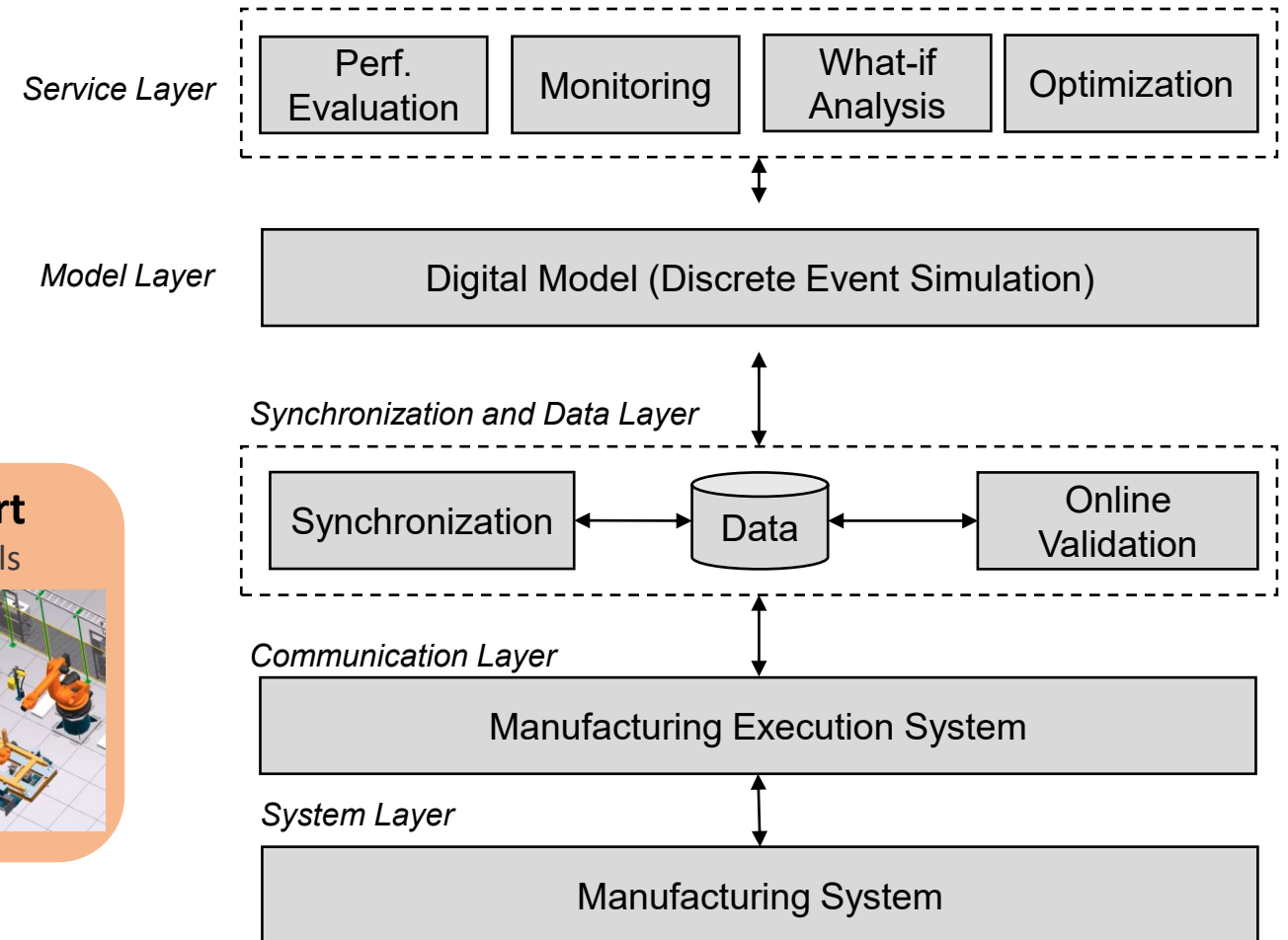


Digital Twin Demonstration

REFERENCE DIGITAL TWIN FRAMEWORK [1]



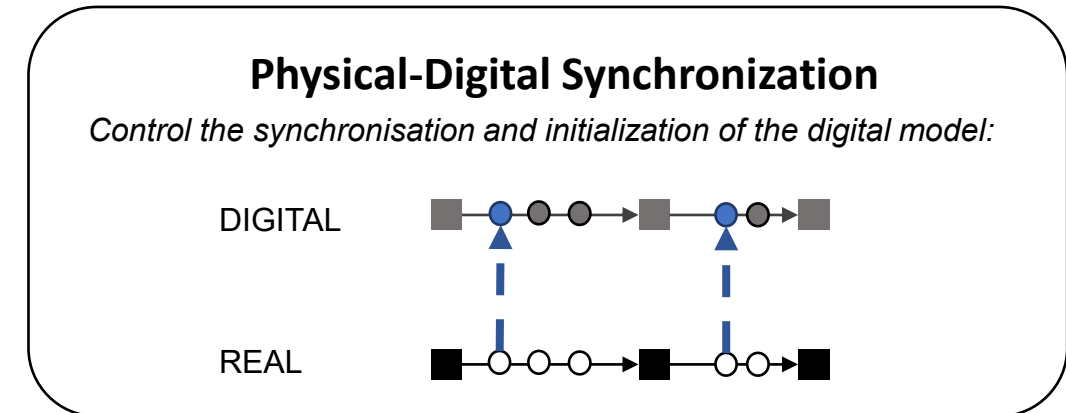
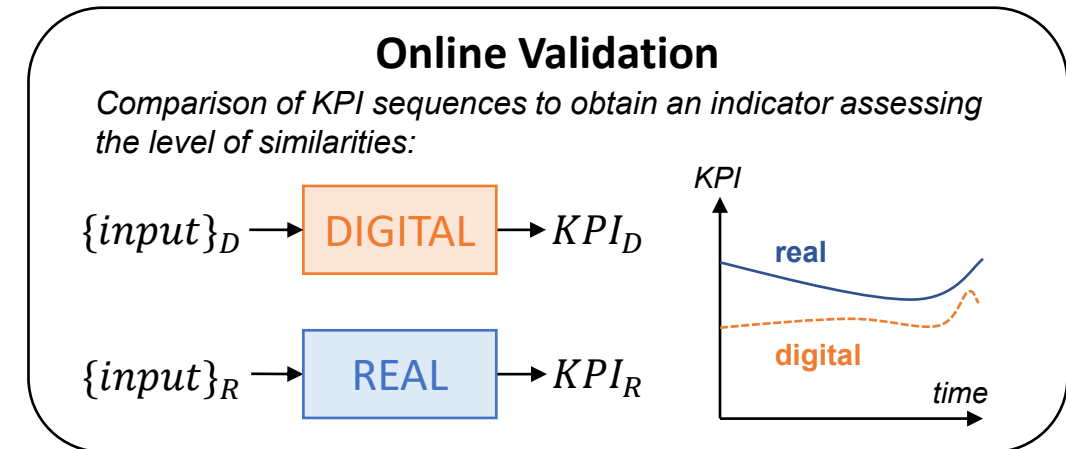
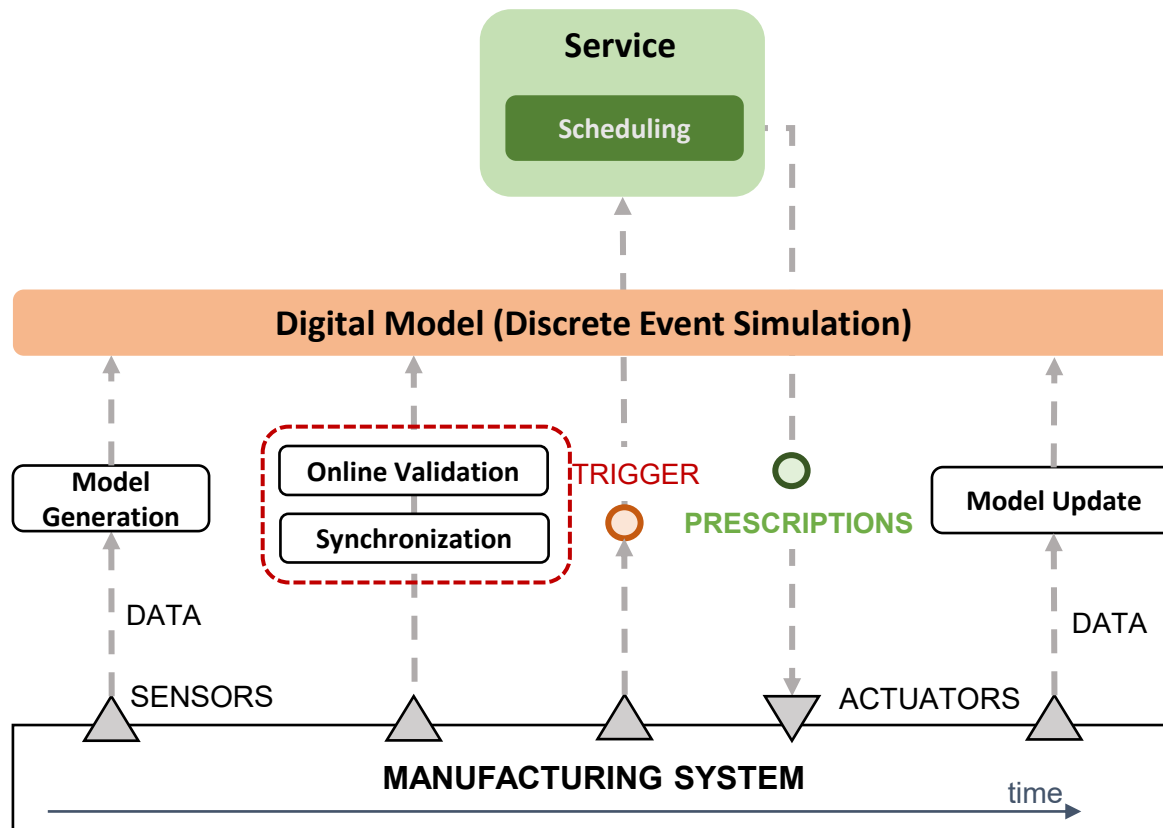
IMPLEMENTATION OF ITS COMPONENTS (architecture)



[1] Tao, Fei, et al. "Digital twin in industry: State-of-the-art." IEEE Transactions on industrial informatics 15.4 (2018): 2405-2415.

Internal Services: Synchronization and Validation

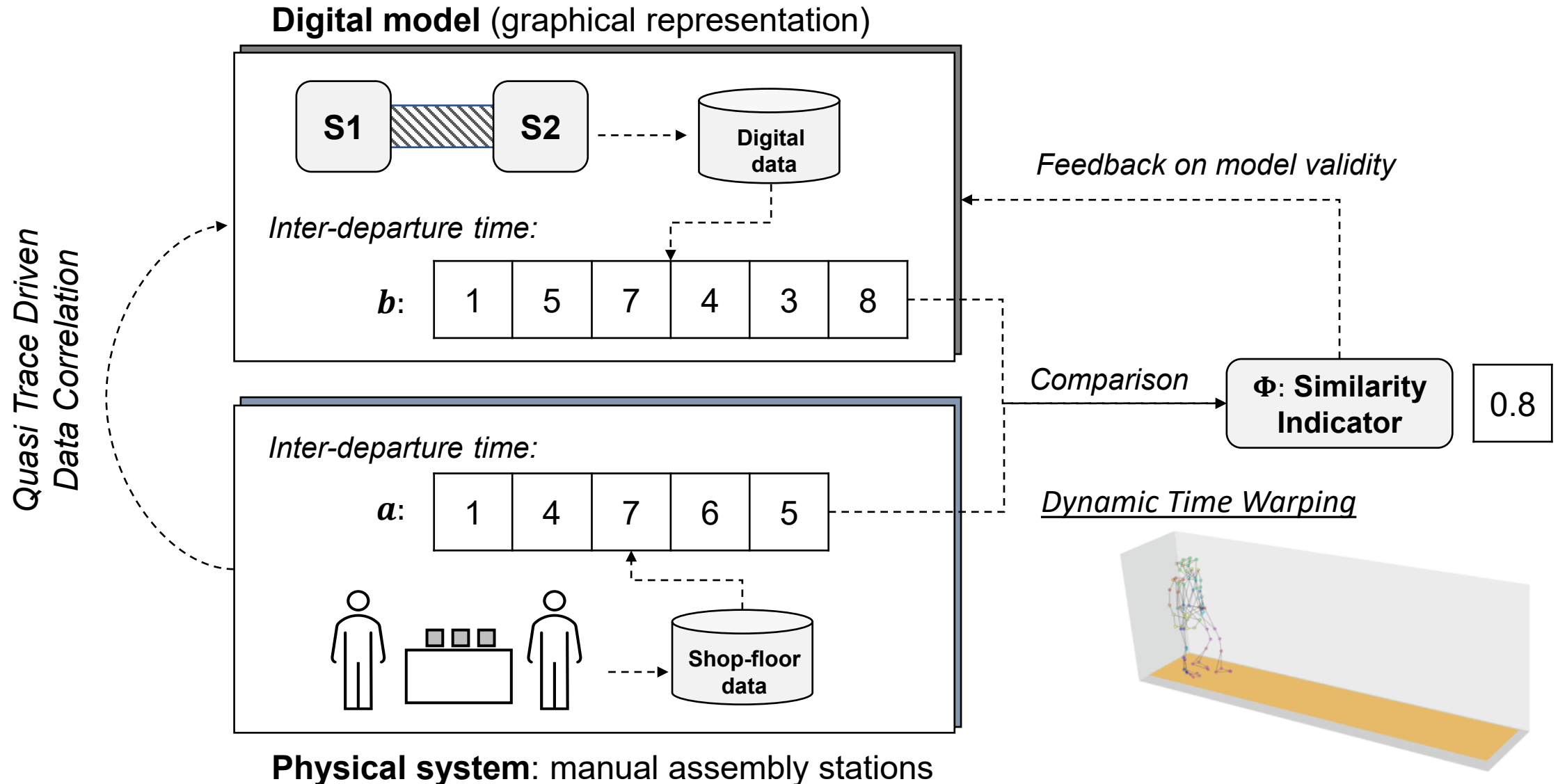
Starting from a specific system state, a digital model may be (1) **generated** from available data, (2) **used online** to provide a particular service, (3) being **updated** to adapt to the changes of the physical system.



[1] "Development of a framework for real-time digital twin alignment". Edoardo Passarin, Francesco Verucchi. M.Sc. Thesis. Supervisor: Prof. Andrea Matta. Co-Supervisor: Giovanni Lugaresi.

[2] "Real-time validation in a digital twin framework". Giulia Gazzoni, Sofia Gangemi. M.Sc. Thesis. Supervisor: Prof. Andrea Matta. Co-Supervisor: Giovanni Lugaresi.

Online Validation Methodology



Online Validation Methodology

Dynamic Time Warping (DTW)

Given $v_1=[1, 5,7,4,3,8]$ and $v_2=[1,4,7,6,5]$, compute $\overline{v_1}$ and $\overline{v_2}$ as:

$$\overline{v_1} = \frac{v_1}{\max(\max(v_1), \max(v_2))} \text{ and } \overline{v_2} = \frac{v_2}{\max(\max(v_1), \max(v_2))}$$

Use the **DTW algorithm** to compute the cumulative distance $dDTW(\overline{v_1}, \overline{v_2})$. Normalize the cumulative distance dividing it by the maximum length of the two sequences.

$$\overline{dDTW}(v_1, v_2) = \frac{dDTW(\overline{v_1}, \overline{v_2})}{\max(m, n)} = \frac{1}{6} = 0.1667$$

DTW indicator is obtained subtracting 1 to the normalized cumulative distance.

$$DTW(v_1, v_2) = 1 - \overline{dDTW}(v_1, v_2) = 0.8333$$

DTW-based indicator matrix

	$\overline{v_1}$	0.125	0.625	0.875	0.5	0.375	1
$\overline{v_2}$	0	∞	∞	∞	∞	∞	∞
0.125	∞	0	0.5	1.25	1.625	1.875	2.75
0.5	∞	0.375	0.125	0.5	0.5	0.625	1.125
0.875	∞	1.25	0.375	0.125	0.5	1	0.75
0.75	∞	1.75	0.5	0.25	0.375	0.75	1
0.625	∞	2.25	0.5	0.5	0.375	0.625	1



$$0.125 = (0.625 - 0.5) + \min\{0.375, 0, 0.5\}$$

- Best alignment
- Cumulative distance

Digital Twin Test Laboratory

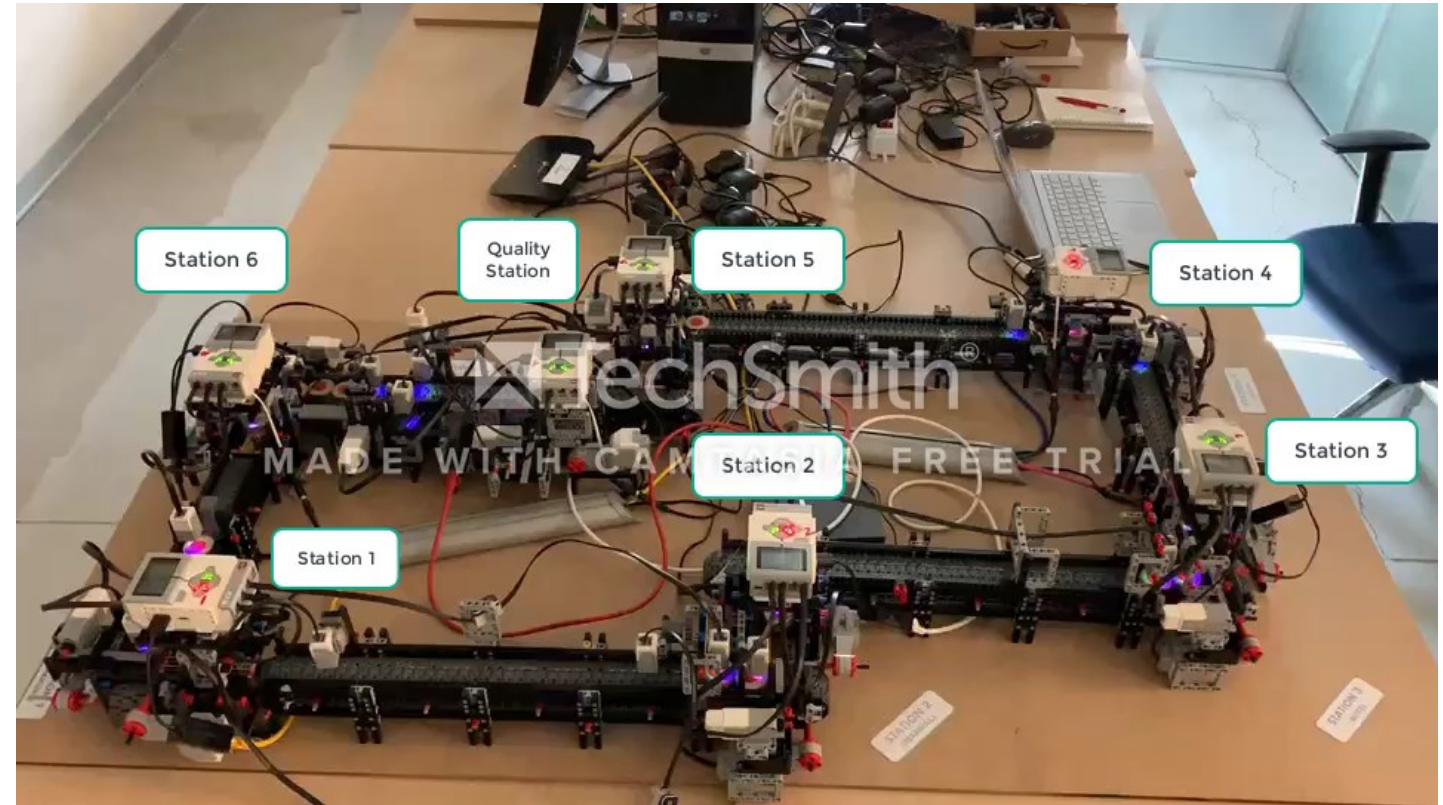
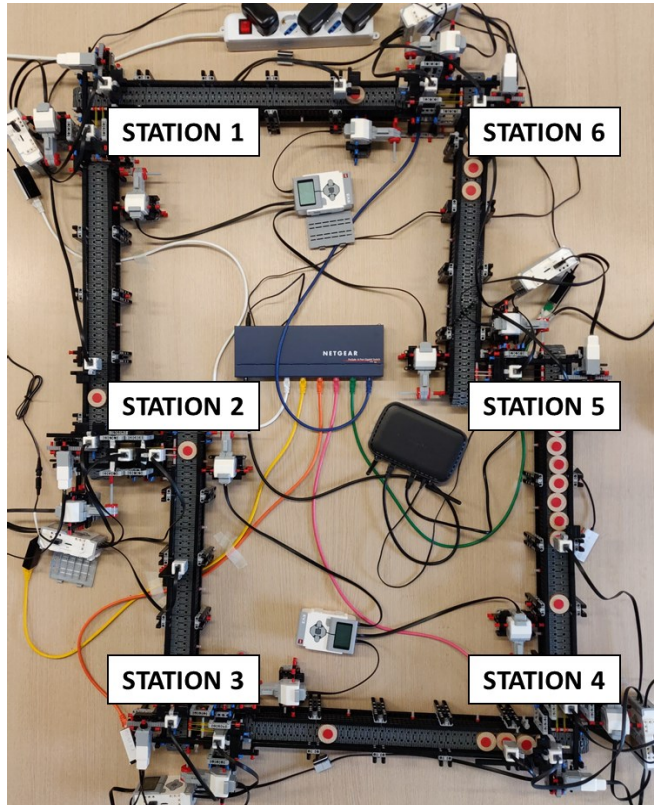
Link to the lab trailer:

https://www.youtube.com/watch?v=G_IvFuYTugE&ab_channel=GiovanniLugaresi



New Laboratory for testing Real-Time Simulation based on the needs from the literature and considering I4.0 developments (e.g., Internet of Things, Cyber Physical Systems).

- Stations are controlled by LEGO® EV3® intelligent bricks.
- Each station has sensors for entrance checking, processing, blocking.
- Wooden circles tagged with red plates represent pallets/parts.

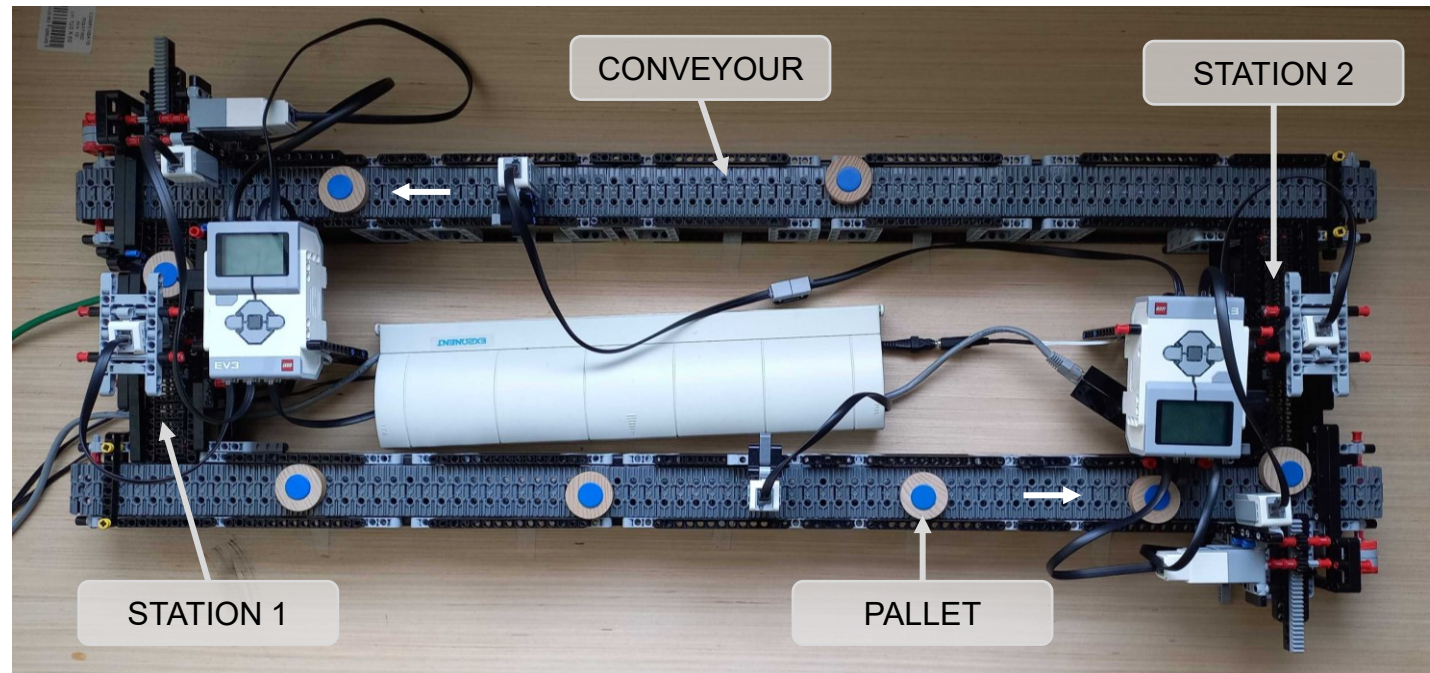


Lugaresi, G., V. Alba, and A. Matta. "Lab-scale models of manufacturing systems for testing real-time simulation and production control technologies." *Journal of Manufacturing Systems* 58 (2021): 93-108.

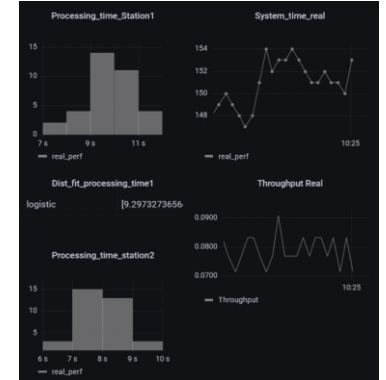
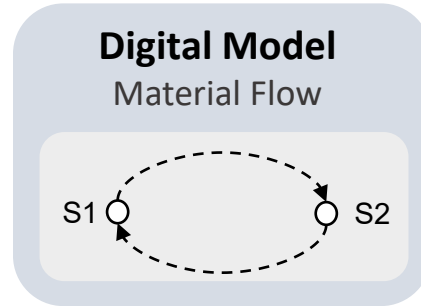
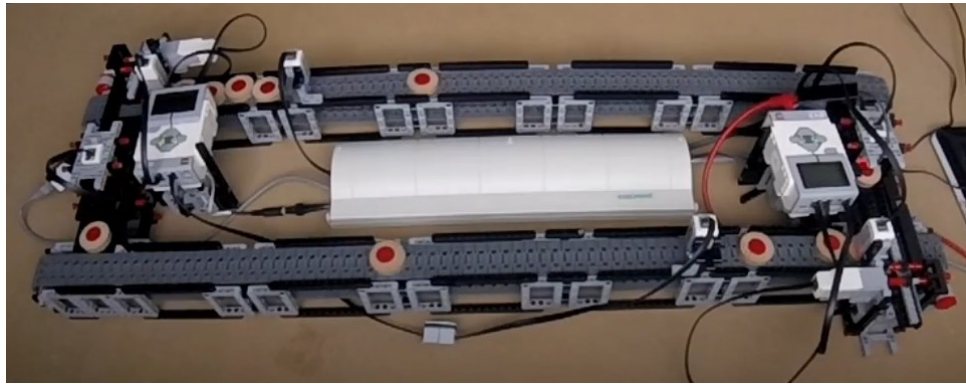
Demo 1: Online scenario-evaluation

A physical system with these characteristics:

- 2-station closed-loop system composed
- Conveyors that operate also as buffers.
- Fixed amount of 12 pallets
- Reliable stations with processing times: (1) triangular distribution: [3, 8, 5] s, (2) triangular distribution: [2, 5, 3] s



Demo 1: Online Scenario-evaluation



Start Physical System
Automated data collection

Digital Model Generation
(automated procedure)

Start Monitoring and
Physical- Digital Alignment

Disruption

Tool degradation is
manifested in a
processing times
distribution change

Model Validation
automated
detection of the
disruption

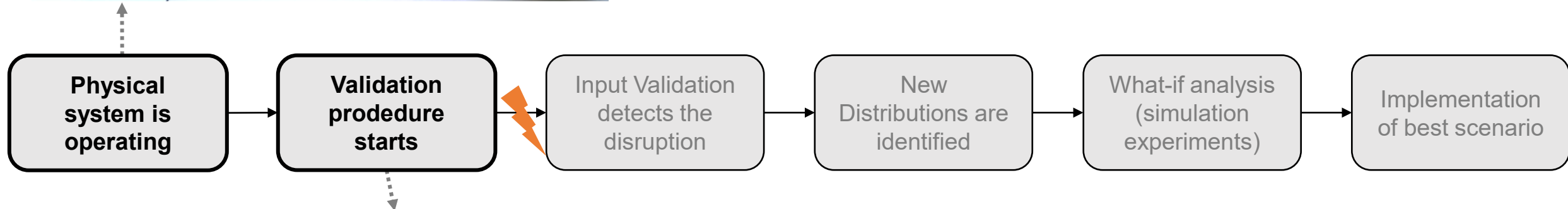
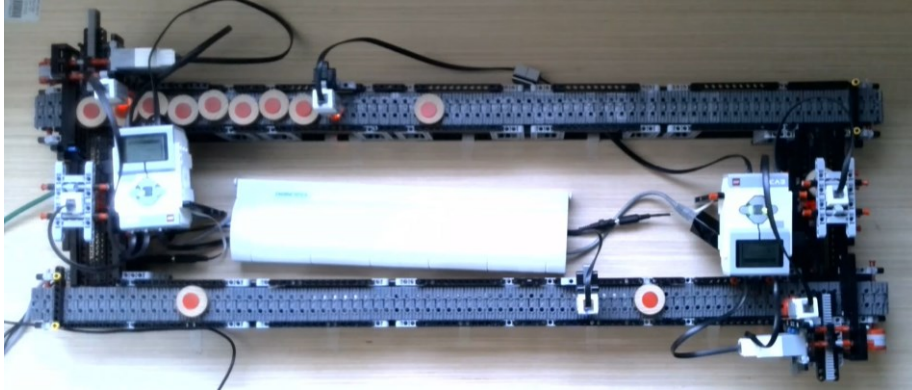
Model Update
New distributions
fitting

Scenario Analysis
Different options are
evaluated in the
digital model

Application of the
best scenario

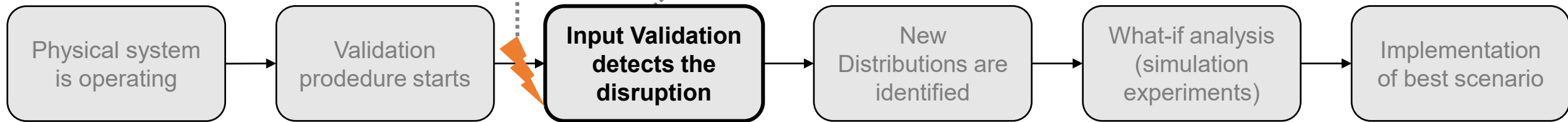
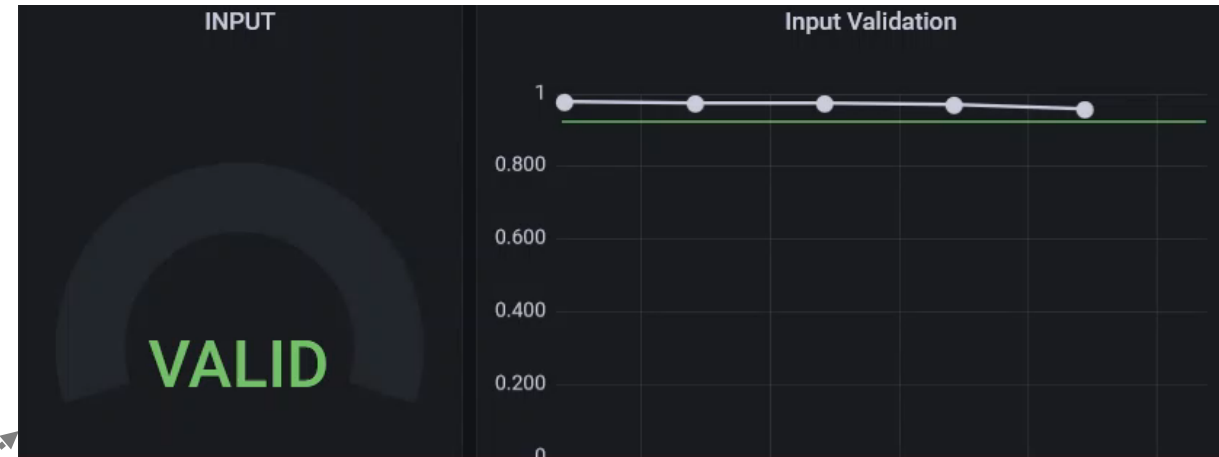
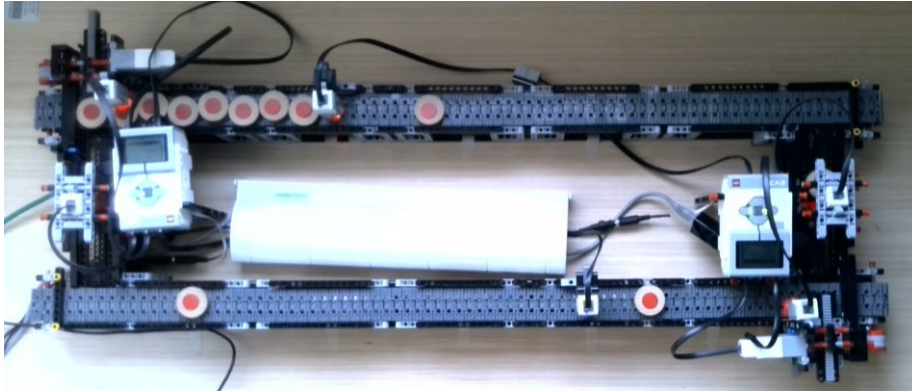
Demo 1

- Reference KPI: System time
- Query data frequency: *1 min*
- Query size: 10 minutes (~30 data points)
- Acceptance threshold: *0.92*



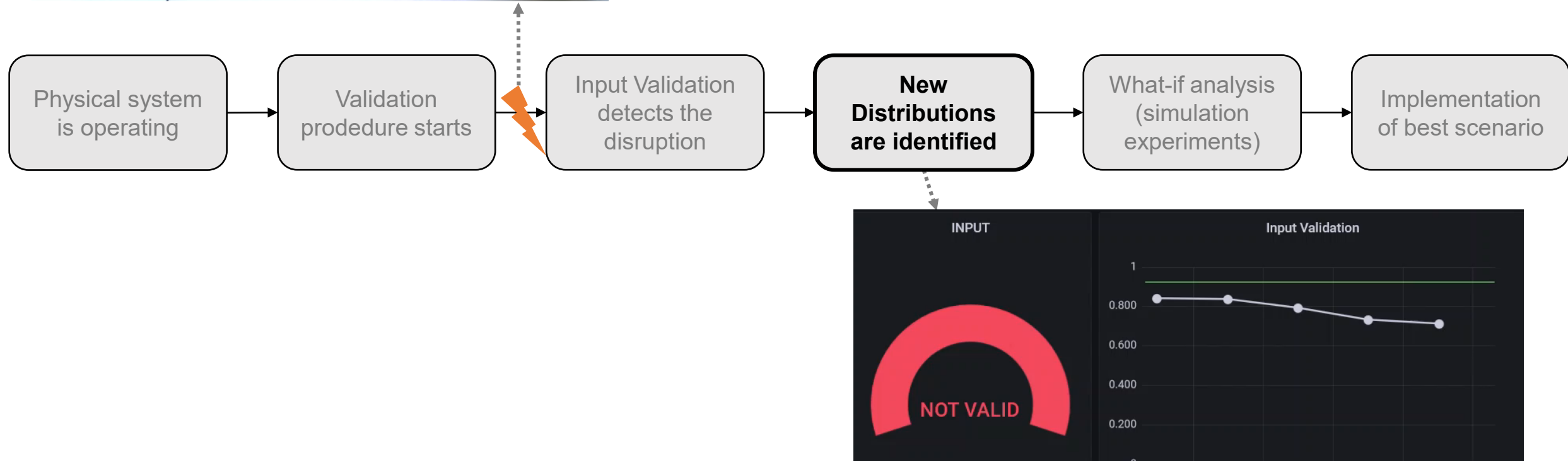
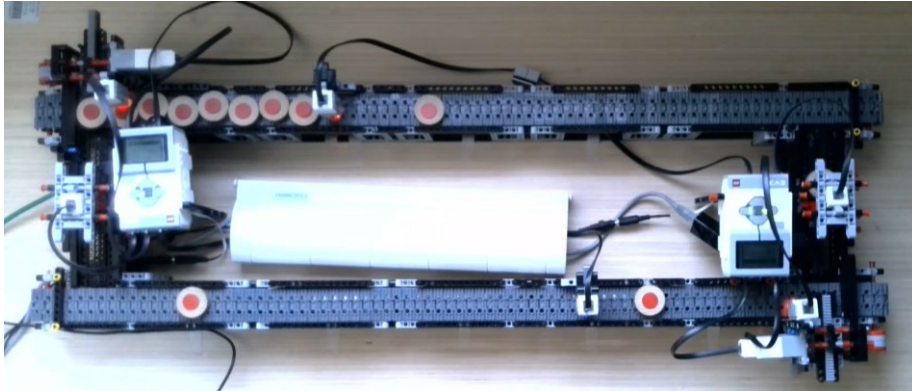
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Demo 1

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- Query data frequency: *1 min*
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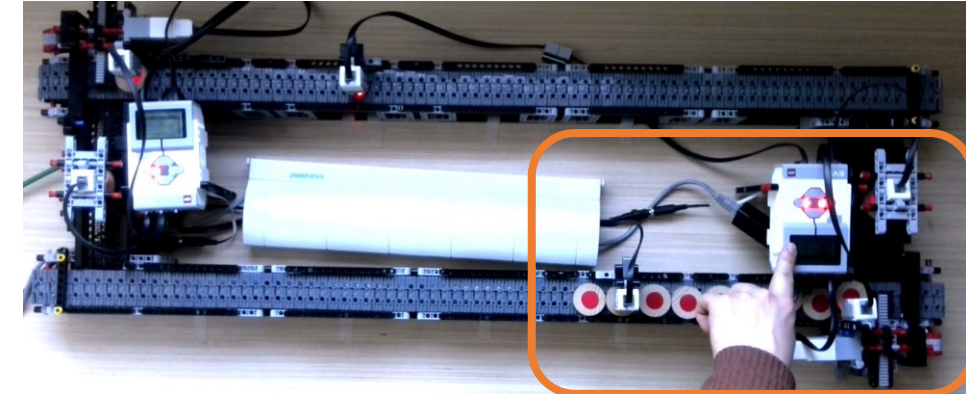
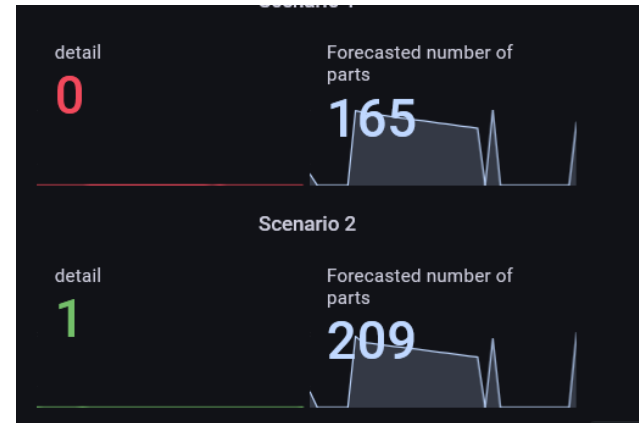


Demo 1

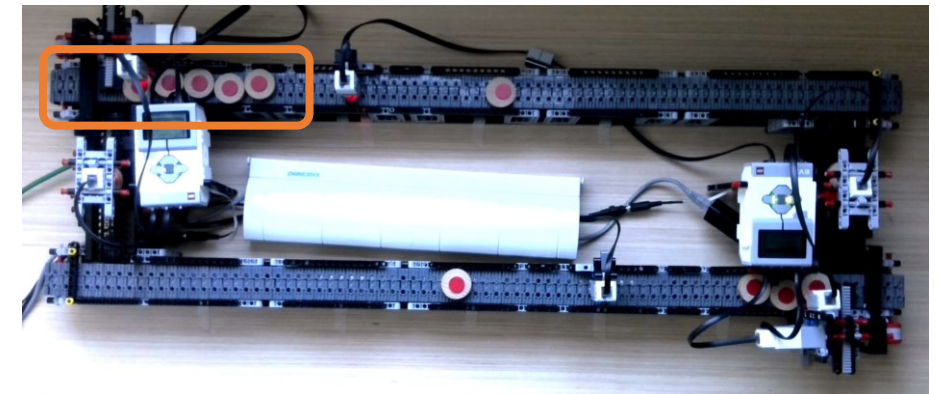
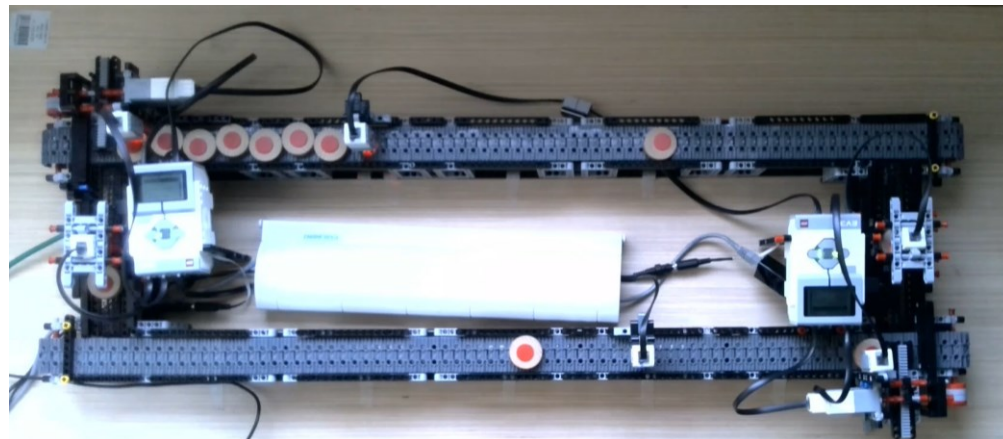
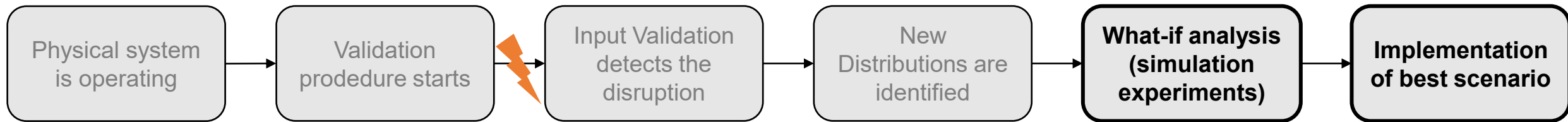
- Reference KPI: System time
- Query data frequency: 1 min
- Query size: 10 minutes (~30 data points)
- Acceptance threshold: 0.92

Scenario analysis

- Two options are evaluated in the digital model
- The **second scenario** is expected to produce more at the end of the time horizon.



The second station is **repaired** (automated decision)

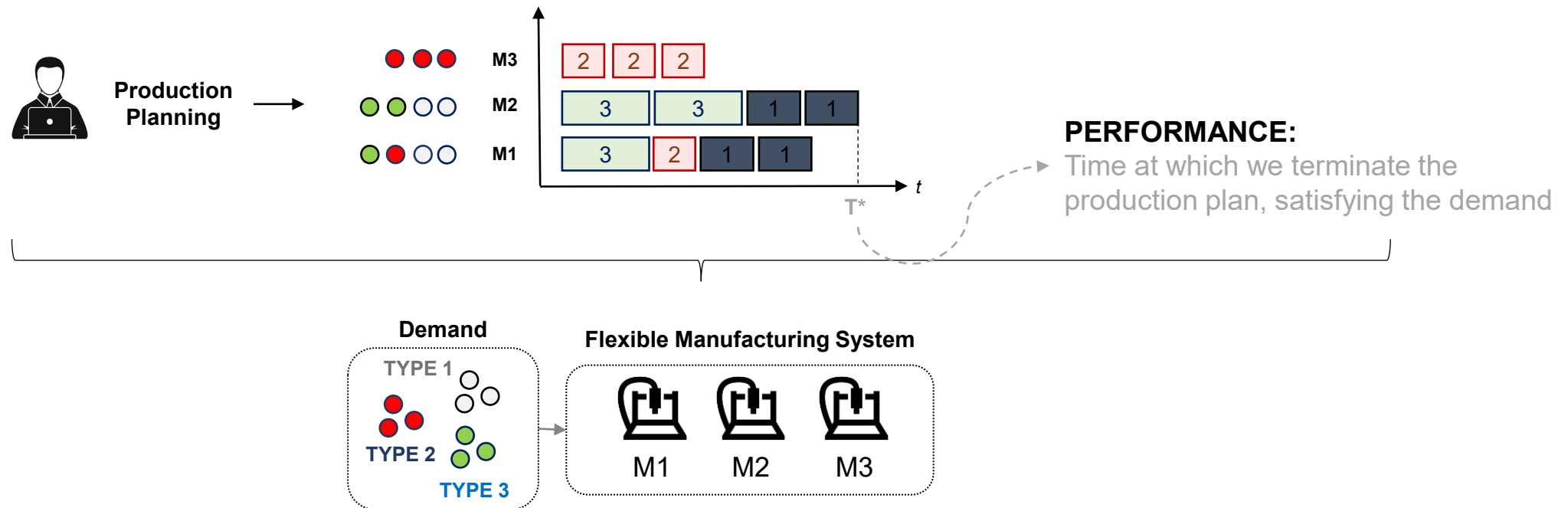


The system goes **back to production** with original parameters

Demo 2: Re-scheduling

Setting of a test in the laboratory:

- A production system is composed by three identical machines.
- Three product variants are being produced on this system.
- A good production plan shall satisfy the demand in the shortest time possible.

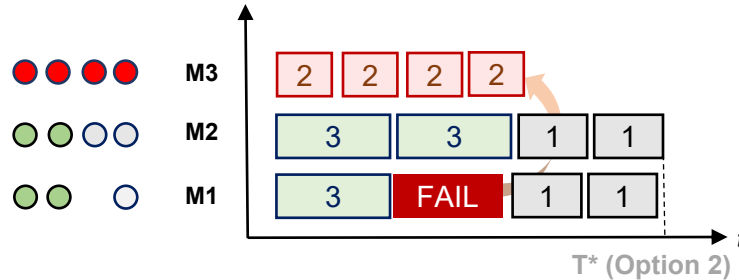


Demo 2: Re-scheduling

Results of the tests in the laboratory:

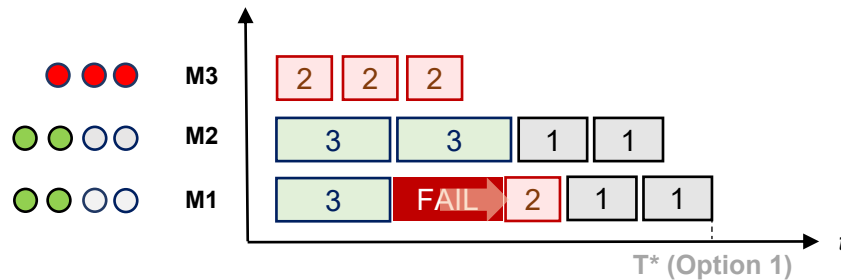
OPTION 2

Re-schedule.



OPTION 1

Shift the production plan on the first machine.



SIMULATION



How to decide?

SIMULATION

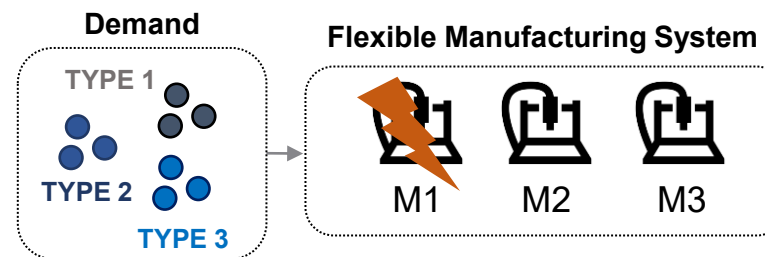
PERFORMANCE MEASURE

Time at which we terminate the production plan, satisfying the demand

$T^* = 363.2 \text{ s}$

$T^* = 404.9 \text{ s}$

10 %
IMPROVEMENT



The best option is applied on the system.

Scenario: Machine 1 fails after 1min-40s for a downtime of 60s.

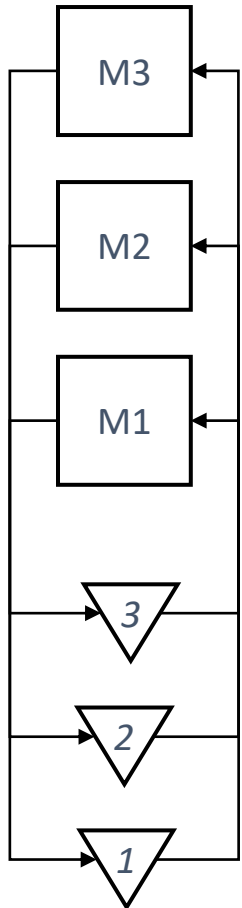
Demo 2: Re-scheduling

Link to the video:
<https://youtu.be/UWq0fn25Xw>

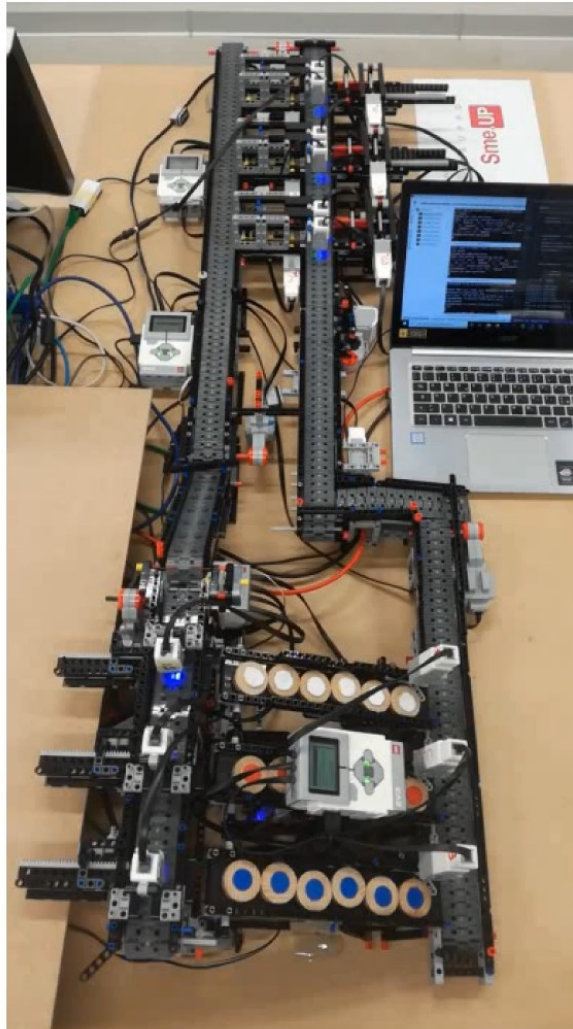


Video of the test in the POLIMI laboratory:

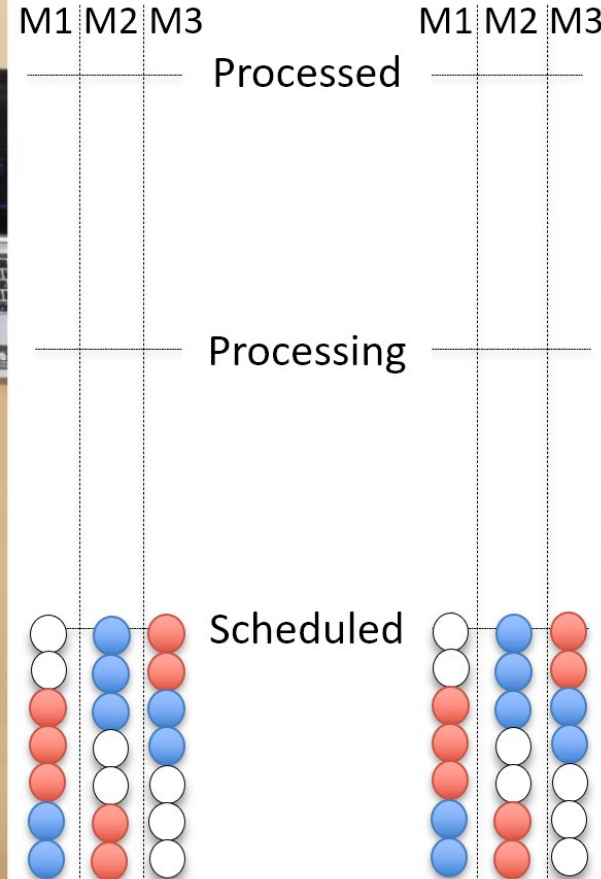
SYSTEM



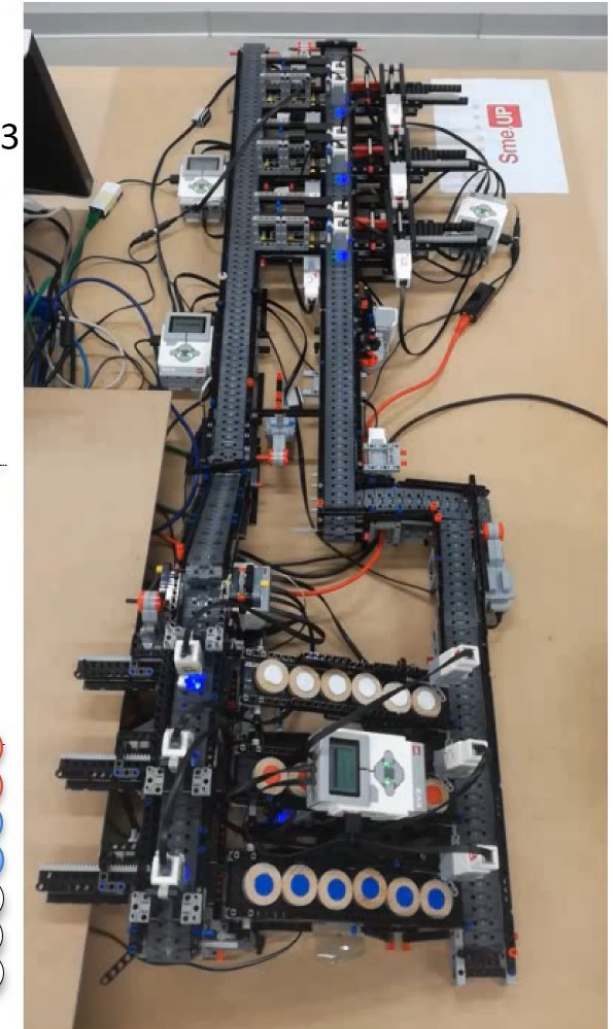
ONLINE RESCHEDULING



00:00



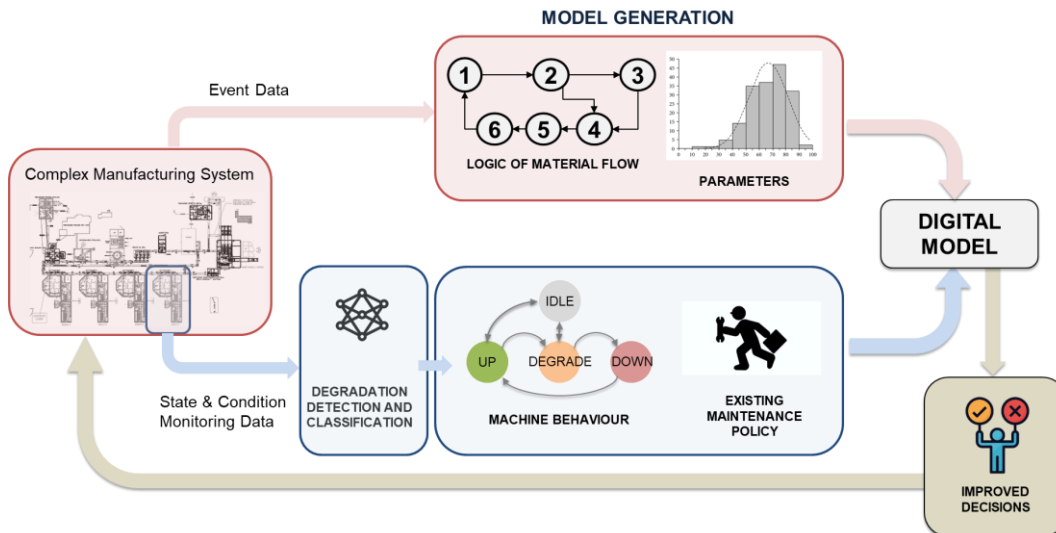
STATIC REACTION RULE



Conclusions and Future Developments

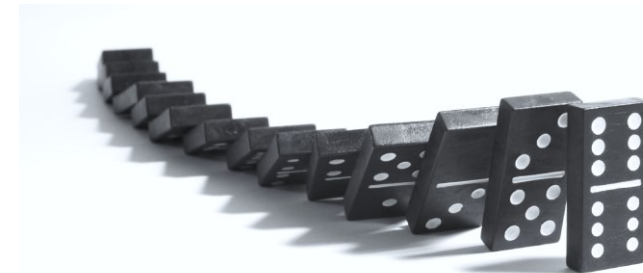
FUTURE DEVELOPMENTS

- Development of **joint mining approaches** → discovery of features of the system (e.g., maintenance policies)
- Using process mining to **enhance performance estimation** → support for approaches that need a model to estimate performance under specific scenarios.
- Extension of **model conversion techniques** → connectors to software tools
- Collaboration with **interested industrial partners** to test the approaches within realistic settings (e.g., MES integration).
- Development of **online validation techniques** to «close the loop» in real-time.



RESEARCH IMPACT

- Digital twins as **online decision-support tool**
- Digital Manufacturing: closer to **real applications of digital twins**
- **New ways to test** softwares and PPC technologies



Research Opportunities (Model Generation)

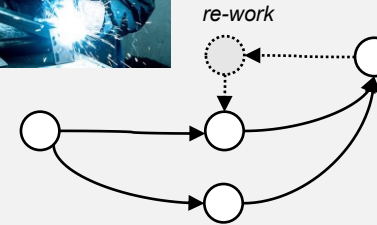
Inclusion of the manufacturing system **properties**



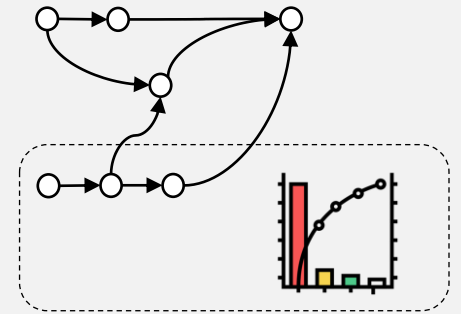
Control the **level of detail** of the model



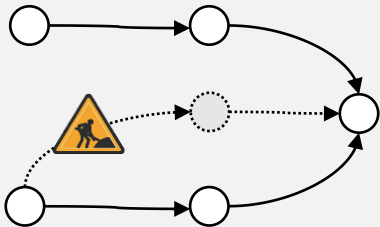
Properly manage **rare events**



Relate model generation with performance estimation



Take into account the manufacturing system **state**



Include additional production control policies (e.g., queue priority rules)

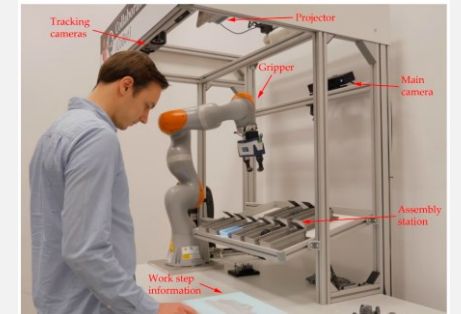
FIFO vs LIFO



Model multiple types of material flows in the same representation



Inclusion of manual operations (i.e., unstructured data)



Q&A and Contact Details

KU LEUVEN

Selected References

G. Lugaresi and A. Matta. Real-time simulation in manufacturing systems: Challenges and research directions. 2018 Winter Simulation Conference (WSC), pp. 3319–3330, IEEE.

Lugaresi, Giovanni, et al. "Real-time validation of digital models for manufacturing systems: A novel signal-processing-based approach." 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE). IEEE, 2019.

R. Sargent. Verification and validation of simulation models. Journal of Simulation, 7:12–24, 2013.

M. Camargo, M. Dumas, and O. González-Rojas. Automated discovery of business process simulation models from event logs. Decision Support Systems, 134:113284, 2020.

A. Khan, M. Dahl, P. Falkman, and M. Fabian, "Digital twin for legacy systems: Simulation model testing and validation," in 2018 IEEE 14th International Conference on Automation Science and Engineering (CASE), pp. 421–426, IEEE, 2018



 EMAIL



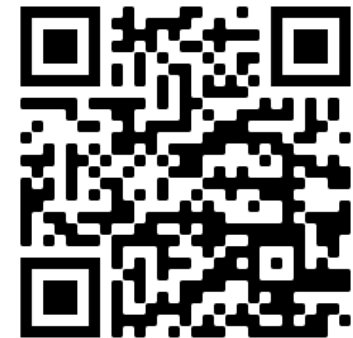
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Automated Generation and Operation of Digital Twins for Manufacturing Systems



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